The Annals of Applied Statistics
2023, Vol. 17, No. 1, 476–497
https://doi.org/10.1214/22-AOAS1638
© Institute of Mathematical Statistics, 2023

ESTIMATING PROMOTION EFFECTS IN EMAIL MARKETING USING A LARGE-SCALE CROSS-CLASSIFIED BAYESIAN JOINT MODEL FOR NESTED IMBALANCED DATA

BY SABYASACHI MUKHOPADHYAY[1,a], WREETABRATA KAR[2,b] AND GOURAB MUKHERJEE[3,c]

[1]*Department of Quantitative Methods and Information Systems, Indian Institute of Management,* [a]*sabyasachi.mukhopadhyay@iimu.ac.in*

[2]*Krannert School of Management, Purdue University,* [b]*wkar@purdue.edu*

[3]*Department of Data Sciences and Operations, University of Southern California,* [c]*gourab@usc.edu*

We consider a large-scale, *cross-classified nested* (CRON) joint model for modeling customer responses to opening, clicking, and purchasing from promotion emails. Our logistic regression-based joint model contains crossing of promotions and customer effects and allows estimation of the heterogeneous effects of different promotion emails, after adjusting for customer preferences, attributes, and historical behaviors. Using data from an email marketing campaign of an apparel retailer, we exhibit the varying effects of promotions not only based on the contents of the email but also across the three different stages, viz. open, click, and purchase of the conversion funnel. We conduct Bayesian estimation of the parameters in the joint model by using a block Metropolis–Hastings algorithm that not only incorporates nested subsampling to tackle the severe imbalance between conversions and no conversions but also uses additive transformation-based modifications of random walk Metropolis to scale estimation for large numbers of customers. We extend our approach to a *segmented cross-classified nested* (SCRON) joint model that encompasses the possibility of varying promotion effects across different customer segments. The resultant high-dimensional model is estimated using spike-and-slab priors on the promotion and customer segment interactions. Our nested joint model accounts for the correlations in customer preferences across the conversion funnel. Based on the promotion estimates from the model, we demonstrate how marketers can use different priced, nonpriced, and combination of price and nonprice promotions to increase brand awareness or increase purchases. Comparing estimates from CRON and SCRON models, we display the benefits of targeted marketing by using email promotion lists which are separately optimized for the different customer segments.

1. Introduction. Over the past decade the digital revolution has changed the paradigm of marketing promotions. According to studies by Forrester Research and eMarketer, in 2020 an average firm was expected to allocate 45% of their total marketing budget to online media (Leone (2020)). Among the different types of digital promotions, email marketing is the most popular, as it yeilds the highest *return on investment* (ROI) (Lee (2019)). The massive popularity of email marketing is primarily due to two factors: its reach, as at least 91% of current internet users participate in digital activities using emails, and second, the minimal cost of sending out an email compared to other digital media (Perrin (2019)). Despite such wide usage, promotion campaigns through emails are currently facing the colossal challenge of relevance. In a recent study by Aldighieri (2019), it was reported that, while 55% of marketers

Received April 2021; revised April 2022.
Key words and phrases. Logistic regression, joint modeling, imbalanced data, spike-and-slab priors, Bayesian hierarchical model, promotion email, high-dimensional regression.

think that more than half of all the emails sent to customers were relevant, only 14% of customers believe that more than half of the emails from an organization were relevant to them. Promotions lacking relevance cause customer alienation and fatigue that may eventually lead to unsubscription of emails or even substitution by competitors (Forte (2019)). Hence, it is extremely important to understand the role of different promotions so that relevant, refined, and targeted email campaigns can be launched.

Marketers are increasingly using data-driven techniques to study promotion effects in email campaigns (see Sahni, Wheeler and Chintagunta (2018), Sahni, Zou and Chintagunta (2017), Zhang, Kumar and Cosguner (2017) and the references therein). Here, based on customer responses to a retailer's email marketing campaign (which is the application case described in detail in Section 2), we develop a disciplined statistical framework to accurately estimate varied promotion effects. Promotions can be classified into two fundamentally different groups: (a) *priced*, which includes discount percentages or discount dollars of varied magnitudes, and (b) *nonpriced*, such as buy-one-get-one free or free shipping or free samples or free returns. Along with the above types, modern-day retailers also use different combinations of priced and nonpriced promotions. Statistically, this gives rise to a high-dimensional estimation problem, where the efficacy of multiple promotions simultaneously used in an email campaign needs to be estimated. These new-age problems are further compounded with nonlinearity, imbalance, heterogeneity, and Big Data related computational challenges. For this purpose we develop a flexible and novel Bayesian methodology that conducts efficient estimation in the presence of the aforementioned challenges in a high-dimensional regression model. These features of our method are described in Section 1.3. Next, we briefly describe the chief objectives and the associated responses in digital marketing campaigns.

1.1. *Email marketing*: *Promotions and conversion funnel*. Most contemporary email marketing campaigns track their effectiveness by evaluating three metrics: open, click, and purchase rates of promotion emails. The open-click-purchase metrics are not mutually independent but are interdependent nested stages constituting the *conversion funnel* (Zhang, Kumar and Cosguner (2017)), which a customer starts by opening the promotion email. These metrics are highly heterogeneous across different campaigns. For the U.S. market the open rates of email promotions in 2018 varied between 19% and 54%, while the click rate varied between 3% and 10% (Perrin (2019)). Purchase, being the terminal point at the bottom of the funnel, is directly related to revenue. However, it is equally important to estimate a promotion's effectiveness at the open and click stages, for they reflect brand awareness, customer acquisition, and customer engagement, which are important attributes for tracking daily operations and business growth. Different promotion strategies have disparate impacts on the aforementioned three stages of the conversion funnel. For estimating promotion effectiveness, we develop a joint modeling framework (Rizopoulos (2012)) that incorporates the interdependencies among the customer responses across the conversion funnel.

1.2. *Joint modeling of customers' responses to promotion emails*. We model customer responses to an email marketing campaign of an apparel company. Finding relevance of different kinds of promotions in this sector is particularly important, as, on average, 60% of shoppers currently feel that promotion emails do not cater to their tastes and interests (Kapner (2017)). We use a nested logistic regression setup for the three stages in the conversion funnel. Recently, Zhang, Kumar and Cosguner (2017), using a hidden Markov model, showed that it is critical to control for the correlated nature of the conversion funnel. However, they considered an aggregated setup where conversion denoted one or more purchases made by a customer from any number of emails she might have received in a month. Thus, their model is agnostic about the promotion content. Our joint modeling framework, based upon a generalized linear model (GLM), not only allows for modeling co-dependencies across the three

stages in the conversion funnel but also allows for measuring customer responses at the disaggregative level for all promotion types and contents. It can generate rich insights into which promotions are more effective at what stage of the conversion funnel, thereby developing actionable insights for managers, depending on what the objective of the promotion is. To the best of our knowledge, this ranks among the first works on joint modeling of the conversion funnel based on promotion content types.

Our proposed GLM-based joint modeling framework predicts customer choices based on promotion effects as well as customer's recency, frequency, and monetary (RFM) values. We entertain nonlinear effects of a customer's recency values, measured in days since she made her last purchase or days since her latest clicking or opening of a promotion email. These RFM variables are described in Section 2. A critical challenge in jointly modeling the three stages is high imbalance between the positive and negative responses, because of low click rates and even lower purchase rates in promotion emails. We develop an imbalance-corrected Bayesian logistic regression framework that also encompasses interactions between RFM values and promotion effects. The resultant model can be used in designing targeted marketing campaigns (as done in Section 5.4 of the application case) that use different promotion contents for different segments of the customer base.

1.3. *Background and statistical challenges*. Traditionally, the effectiveness of promotions is studied at the end stage of the conversion funnel based on buy or not buy decisions. While Kamakura and Kang (2007) and Osinga, Leeflang and Wieringa (2010) used sale response model to study purchases at the aggregate level, recent works of Wu, Li and Liu (2018), Zhang, Kumar and Cosguner (2017) demonstrated the benefits of conducting efficiency analysis at the individual customer-identifier level. Using randomized field experiments, Gopalakrishnan and Park (2021) and Sahni, Wheeler and Chintagunta (2018) demonstrated empirical evidence of the heterogeneous behavior of customers across the different stages of the conversion funnel. Here, we use Bayesian joint modeling (McCulloch (2008), Rizopoulos (2012), Rizopoulos and Ghosh (2011), Rizopoulos and Lesaffre (2014)) which provides a disciplined framework for simultaneously studying customer-level responses across the three stages of the conversion funnel. However, email-marketing data containing responses to multiple email promotions from a large number of customers, and the business model associated with marketing funnels poses several statistical challenges and necessitates novel extensions of the conventional Bayesian joint modeling regression framework. We describe the salient features of our approach below.

Evaluating promotion content and a two-way cross classified joint model. The existing literature (see Seetharaman and Chintagunta (2003), Wu, Li and Liu (2018), and the references therein) mainly focuses on determining the optimal promotion frequency and volume and very limited research has delved into the content of a digital promotion. The extent of research on promotion content has been mostly restricted to determining, given a customer segment(s) or engagement propensity, whether a price promotion was better than a nonprice promotion. However, given the current variation in offers, the above insight is not quite actionable for a retailer. The retailer needs to know whether a price promotion should be about discount percentage or discount dollars; whether a nonprice promotion should be buy-one-get-one free, or free shipping, or free samples, or free returns, and how much are the differences between their effects, if any; whether it is beneficial to use combinations of priced and nonpriced promotions. For that purpose, here we estimate the efficacy of each of the different promotion emails (different subject line or text content) used in the campaign. Statistically, this translates to considering two-way cross-classified models with crossed effects for customers' personal preferences to email marketing and a promotion email's attractiveness.

We use correlated random effects for modeling individual customer preferences at the different stages of the conversion funnel, and fixed effects for studying her responses to promotions. While using a cross-classified model enabled us to estimate promotion effects, we encounter severe imbalance in the responses (between conversion and no conversion), which is controlled in aggregated models that does not differentiate between promotion contents as in Zhang, Kumar and Cosguner (2017), who aggregated responses at the monthly resolution. Also, estimating parameters in a large cross-classified model with random effects is challenging. We describe these issues next.

Imbalanced data. As the conversion rate in email marketing can be quite low, often severe imbalance is observed in the purchase stage of the conversion funnel. It is well known (see Chapter 16 of Kuhn and Johnson (2013), Fithian and Hastie (2014), Johndrow et al. (2019), and the references therein) that, in modeling discrete classes, highly disproportional relative frequencies of the classes can have a significant impact on the effectiveness of the model. Subsampling approaches with subsequent correction of the regression estimates (Fithian and Hastie (2014), Wang, Zhu and Ma (2018)) is one of the popular approaches to tackling imbalance. Here, in Section 4.2.1, we develop a nested subsampling scheme that can correct imbalance across the different stages of the conversion funnel. In our application (described in Section 2), customer responses in purchase stage was severely imbalanced whereas those in the open and the click stages were not disproportionately imbalanced. So, we applied the algorithm with subsampling only in the purchase stage.

Scalable estimation for Big Data. Unlike recent works in email marketing (Zhang, Kumar and Cosguner (2017) considered 200 and Wu, Li and Liu (2018) had 2000 focal customers), here we have a very large sample of customers (around 78,000). Estimating regression coefficients in such a massive cross-classified *generalized linear mixed model* (GLMM) presents several computational challenges. As such, scaling mixed effects models in massive customer-level data for modeling the customer's personal preferences to a product is a vibrant topic in contemporary statistics (Gao (2017), Gao and Owen (2017), Zhang et al. (2016)). In a two-way cross-classified balanced linear mixed effect model, Papaspiliopoulos, Roberts and Zanella (2020) have recently established fast convergence of a collapsed Gibbs sampler. We use a related block Metropolis–Hastings (BMH) algorithm (see Section 4.2.2) for Bayesian estimation of the parameters in our large and complex cross-classified joint modeling setup. To scale estimation, we use additive transformation-based modifications of random walk Metropolis (Dey and Bhattacharya (2016), Dutta and Bhattacharya (2014)), which greatly increased the acceptance rate in BMH. We establish convergence properties of the BMH algorithm with additive transformations (BMHT). We run BMHT in a parallel computing framework on multiple machines, each based on a subsample of the data, suitably selected, as explained later in Section 4.2.1, to tackle imbalanced responses. We combine posterior inference from these multiple runs on subsampled data by using the consensus Monte Carlo approach of Scott et al. (2016) and Huang and Gelman (2005).

Nonlinear RFM effects. Customer characteristics and purchases greatly vary as RFM values changes. To estimate the effects of promotion emails, we need to look at residual customer responses after adjusting for the influence of the RFM variables. However, the effect of the RFM variables is not linear (Bult and Wansbeek (1995), Fader, Hardie and Lee (2005)). Figure 3 shows the marginal distribution of the recency variables. We model the nonlinear effects of these recency variables using cubic splines (see Chapter 6 of James et al. (2013)). However, incorporating a nonlinear structure that not only fits well with the empirical observation but also is tractable in the large-scale joint modeling framework is nontrivial. The flexible generalized cross-validation methodology in Ruppert (2002) would be difficult to implement in our large-scale joint modeling setup. In Section 4.1, based on conditional distributions, we implement a pragmatic, computationally cheap, heuristic policy for capturing nonlinear RFM effects.

Customer segmentation and high-dimensional promotion effects. In the marketing literature it is well documented that promotions may impact different customer segments differently (Gopalakrishnan and Park (2021), Rossi, McCulloch and Allenby (1996)). Assuming that the customer segments are known from previous market research, we introduce interactions between promotions and those customer segments to encompass the possibility of varying promotion effects across different customer segments in our cross-classified joint model. Even with a moderate number of segments, the resultant model is high dimensional. However, most of these interaction effects will not be significantly different from their modal values. In Section 4.4 we use spike-and-slab prior (Castillo, Schmidt-Hieber and van der Vaart (2015), Ročková and George (2018)) on the interaction parameters to discard insignificant effects.

The rest of the paper is organized as follows. In Section 2 we describe our data, which is obtained from email marketing campaigns of an apparel retailer. In Section 3 we present our cross-classified joint modeling framework, and we detail the methodology used for estimating the proposed model in Section 4. Using the data set introduced in Section 2, we present inference and business implications of the results obtained by our proposed analysis in Section 5. We demonstrate how marketers should use different promotions for increasing brand awareness or purchases. We also show (see Section 5.5) the benefits of targeted marketing by using email promotion lists separately optimized for the different customer segments. Figures and tables referred in the paper with the prefix S are presented in the Supplementary Material, which also contains further details on the data set, estimation procedure, and estimated models.

2. Data. Our data set consists of responses from 77,986 customers to a series of promotion emails sent to them in a campaign season. Overall, there were 25 promotion emails that were used during this campaign. Out of these, nine are solely priced (P) promotions, seven are solely nonpriced (NP), and nine are combination (C) promotions having priced as well as nonpriced incentives. Our data contain a recipient's responses to opening, clicking (if opened), and purchasing (if any, after clicking) from each of the promotion emails.

Along with customer responses to the promotion emails, we have the following additional information regarding each customer: (a) *age*, which is binned as a categorical variable with five levels: less than 17 years, 18–35 years, 36–49 years, 50–64 years, and 65 plus years; (b) income, binned as a categorical variable with nine levels: under \$15K, \$15–25K, \$25–35K, \$35–50K, \$50–75K, \$75–100K, \$100–120, \$120–149K, and \$150K plus; (c) three recency variables pertaining to the number of days since the customer last opened (*Days Since Last Opened*), clicked (*Days Since Last Clicked*), and purchased (*Days Since Last Purchased*) from a promotion email; (d) frequency of purchases (Pur. Freq.), tabulated as the number of orders made in the last two years; (e) average order values separately tabulated for in-store (*Avg. Retail Spend*) and internet (*Avg. Web Spend*) purchases. For any customer, who has never purchased from any promotion email, her monetary (Avg. Retail Spend and Avg. Web Spend) and purchase frequency values are set to 0. Her recency (*Days Since Last Purchase*) value is set to the number of days since her email was registered to the marketing database. Similarly, the *Days Since Last Open* and *Days Since Last Click* values are set for any person who never responded to any promotion email.

Figure 1 shows the pairwise histograms of these customer-identifier level attributes; income and age buckets were judiciously clubbed in the figure for presentational ease. Purchase frequency, although a continuous variable, has been converted to categorical variable for representational ease. For each promotion, Table S.2 (in the *Supplementary Material*) presents the aggregated values of these statistics summarized across the recipients, wherein we observe that the average customer characteristics pertaining to the frequency and monetary values (Avg. Retail Spend and Avg. Web Spend) are very similar across all the different

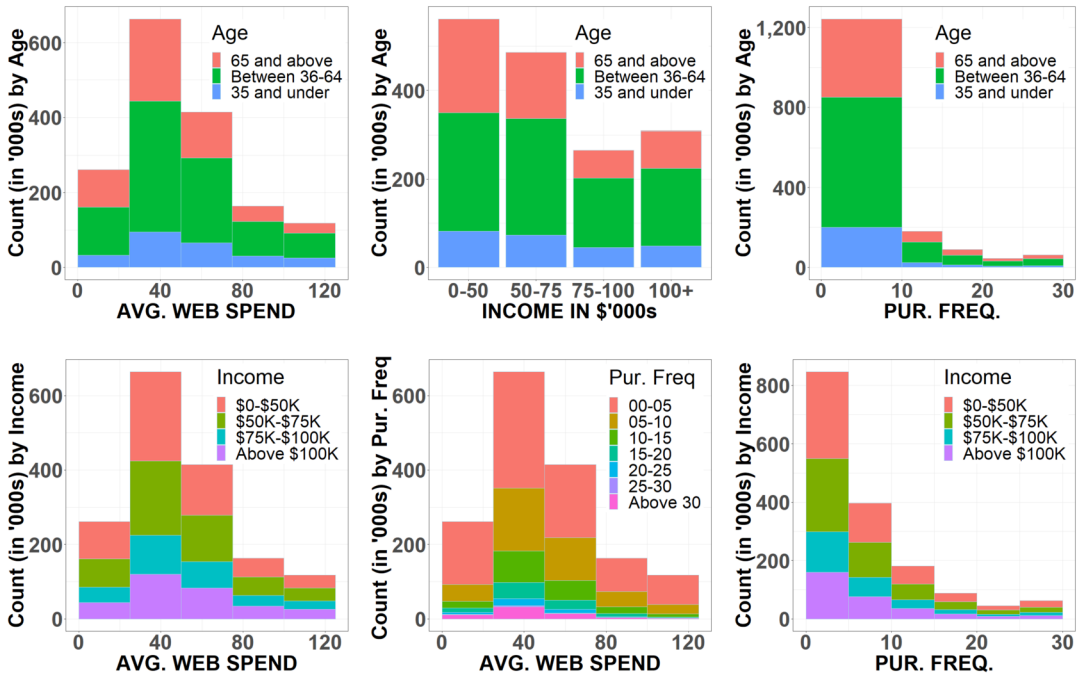


FIG. 1. *Pairwise histograms of customer characteristics*: *Average web spending* (*in USD*), *purchase frequency*, *age* (*in years*), *and income* (*in USD*).

promotions. Figure 2 (right plot) shows the average customer responses for the different promotion emails; the average responses vary significantly, not only across promotions but also across the stages of the conversion funnel.

Figure 2 (left diagram) shows a schematic demonstrating the different components of the data. We model the customer responses at the three different stages of the conversion funnel based on her age, income, recency, frequency, and monetary characteristics as well as the types (P/NP/C) of a promotion email. Every customer has not received all the 25 promotion emails. Table S.1 (in the *Supplementary Material*) shows the distribution of the number of emails received while Figure S.1 (in the *Supplementary Material*) shows the coverage of each of the promotion emails. On average, a customer received around 21 promotions. We were not informed of any targeting in the promotion emails. Using the methodology prescribed in McCaffrey et al. (2013), we compared the balance of each covariate among different promotions (see Table S.3 in the Supplementary Material) and observed strong balance in the covariates. The covariates considered included all the variables that the firm would use to make

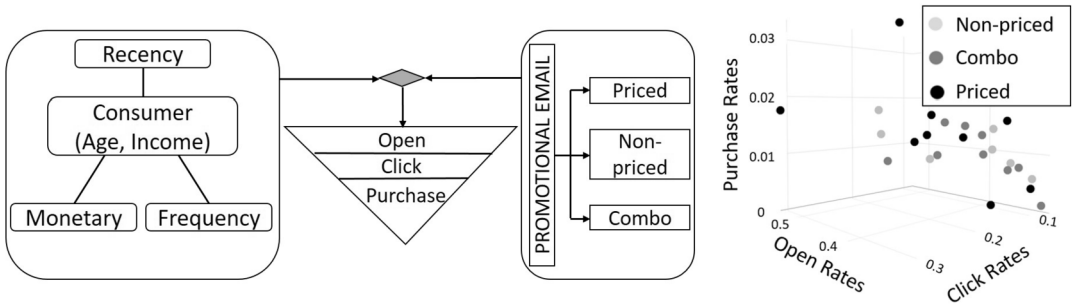


FIG. 2. *Left*: *Schematic diagram showing different components of the data*; *customer responses at different stages of the conversion funnel are jointly modeled based on customer characteristics and the content of a promotion email*. *Right*: *Scatterplot of average customer responses to different promotion emails*.

marketing decisions. In absence of any knowledge regarding the presence of targeting and witnessing strong covariate balance, we assume that the unobserved customer-promotions pairs in our data are missing at random. This randomness assumption is important for unbiased estimation of the models described in the following section.

From the above data, we randomly select $T = 10{,}000$ customers as test data and use the data corresponding to the remaining $N = 67{,}986$ customers as our training data set (for details, see Section S.1 of the Supplementary Material).

3. Model.

3.1. *Cross-classified nested nonlinear model* (*CRON*). Let the binary variables O_{ik}, C_{ik}, and P_{ik} denote the response of customer i at the open, click, and purchase stages, respectively, to promotion email k, where $i = 1, \dots, N$ and $k = 1, \dots, K$; positive values of O_{ik}, C_{ik}, and P_{ik} indicate conversion at the respective stages. Here, $K = 25$. We develop a conditional hierarchical model with its base at the open stage of the conversion funnel. Consider the following two-way cross-classified model:

$$\texttt{logit}\{P(O_{ik} = 1)\} = \boldsymbol{U}_{ik}^T \boldsymbol{\beta}^{(1)} + f_1(\boldsymbol{r}_{ik}) + \nu_k^{(1)} + b_i^{(1)} \tag{1}$$

with $\boldsymbol{b}_i^{(1)}$ being the preferences of customer i and $\boldsymbol{\nu}_k^{(1)}$ being the effect of the kth promotion email. The suffix 1 denotes that it is the model for the first stage (open) of the conversion funnel. $\boldsymbol{\beta}^{(1)}$ is invariant across both promotions and customers. The vector $\boldsymbol{U}_{ik}$ contains the covariates values corresponding to Age, Income, Purchase Frequency, Avg. Retail Spend, and Avg. Web Spend for customer i and promotion email k. Note that, while Age and Income are likely to remain invariant across promotions for a fixed customer, Purchase Frequency, Avg. Retail Spend, and Avg. Web Spend usually vary across k for any fixed i. $\boldsymbol{r}_{ik}$ is the three-dimensional vector of recency variable (last opened, clicked, purchased) that is allowed to have nonlinear effects. In the click stage, note that $C_{it} = 0$ if $O_{it} = 0$, and so we model

$$\texttt{logit}\{P(C_{ik} = 1 | O_{ik} = 1)\} = \boldsymbol{U}_{ik}^T \boldsymbol{\beta}^{(2)} + f_2(\boldsymbol{r}_{ik}) + \nu_k^{(2)} + b_i^{(2)}. \tag{2}$$

Note that $P(C_{ik} = 0) = P(O_{ik} = 0) + (1 - P(C_{ik} = 1 | O_{ik} = 1)) \cdot (1 - P(O_{ik} = 0))$. In this second-stage model, the parameters denote the conditioned (additional) effects, based on the first-stage effects. Similarly, for the purchase stage we consider

$$\texttt{logit}\{P(P_{ik} = 1 | C_{ik} = 1)\} = \boldsymbol{U}_{ik}^T \boldsymbol{\beta}^{(3)} + f_3(\boldsymbol{r}_{ik}) + \nu_k^{(3)} + b_i^{(3)}. \tag{3}$$

Let $\boldsymbol{b}_i = (b_i^{(1)}, b_i^{(2)}, b_i^{(3)})$. We assume $\{\boldsymbol{b}_i : 1 \le i \le N\}$ are independent and identically distributed (i.i.d.) random variables from $N(0, \Sigma)$. Thus, the outcomes O_{ik}, C_{ik}, P_{ik} are allowed to be correlated across the different stages of the funnel.

Figure 3 shows the marginal distribution of the recency variables in the data; the marginal densities are multimodal with greatly varying intermodal distances, suggesting that the recency variables might not only have nonlinear but also local effects. Thus, we use quadratic splines to model those. f_1, f_2, f_3 are modeled by additive splines $f_j(\boldsymbol{r}_{ik}) = \sum_{l=1}^{3} \pi_{jl} f(r_{ikl}; \boldsymbol{\mu}^{(jl)}, \boldsymbol{t}^{(jl)})$, for $j, l = 1, 2, 3$. The weights π_{jl} are binary; some of them were set to zero, based on model diagnostics, to eliminate cases of multicollinearity between the the last opened ($r_{..1}$) and last clicked ($r_{..2}$) covariates. We use quadratic splines in f as

$$f(x; \boldsymbol{\mu}, \boldsymbol{t}) = \sum_{i=1}^{2} \mu_i x^i + \sum_{j=1}^{m} \mu_{2+j} (x - t_j)_+^2, \quad \text{where } m = \text{length}(\boldsymbol{t}). \tag{4}$$

The coefficients ($\boldsymbol{\mu}$), the number of knots ($\boldsymbol{m}$) as well as the location of the knots ($\boldsymbol{t}$) are chosen separately for each stage and each recency variable $r_{..l}$. Henceforth, we call the model in

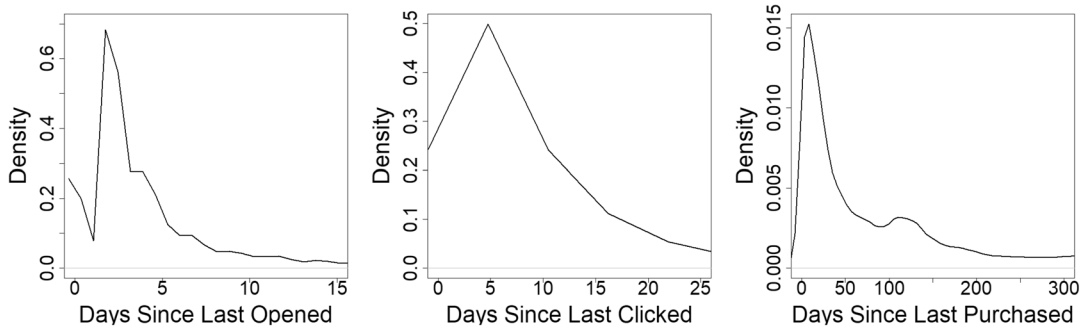


FIG. 3. *Marginal density of the days since last*: (*a*) *opened*, (*b*) *clicked*, *and* (*c*) *purchased*, *respectively*, *from left to right*.

equations (1)–(4) CRON (cross-classified nested nonlinear model). Table S.7 in the Supplementary Material lists its coefficient estimates.

To understand the impact of local effects of recency variables and contentwise promotions in CRON, we compare CRON to the following two nonclassified (NC) nonlinear submodels:

(a) Nonclassified global (NCG), which does not have contentwise promotion but only global effects of the recency variables based on quadratic polynomials. NCG corresponds to imposing the following constraints $\nu_k^{(j)} = \nu^{(j)}$ for all k and $j = 1, 2, 3$, and $\boldsymbol{t} = \varnothing$ in equations (1)–(4).

(b) Nonclassified local (NCL), which has nonlinear local recency variables, as in CRON, but no contentwise promotion effects. Note, that NCG $\subseteq$ NCL $\subseteq$ CRON. Table 1 presents the key features of these models when compared to existing methods.

3.2. *Segmented cross-classified nested nonlinear model* (*SCRON*). We next consider an extension of CRON to encompass scenarios where the promotion effects are allowed to vary with customer type. Suppose the customers are preclassified in $g = 1, \dots, G$ segments and the promotions effects can vary between different segments. Instead of $\nu_k^{(j)}$, we use $\nu_{kg}^{(j)}$ in

TABLE 1
Comparison of SCRON model with existing methods

Model	Customer ID resolution	Joint model of conversion funnel	Locally estimated recency effects	Heterogeneous promo effects based on email content	Varying promo effects on customer segments
Kamakura and Kang (2007)					
Osinga, Leeflang and Wieringa (2010)	×	×	×	×	×
Zhang, Kumar and Cosguner (2017)					
Wu, Li and Liu (2018)	✓	×	✓	×	×
NCG	✓	✓	×	×	×
NCL	✓	✓	✓	×	×
CRON	✓	✓	✓	✓	×
SCRON	✓	✓	✓	✓	✓

equations (1)–(3). As it contains interaction terms between promotion content and customer types, we call it segmented cross-classified nested nonlinear model (SCRON).

Due to these interactions, even for moderate values of G, the GLMs in each stage of SCRON are very high dimensional. However, a lot of the promotion effects $\{\nu_{kg}^{(j)} : k = 1, \dots, K, g = 1, \dots, G\}$ do not differ significantly from the baseline effect. We use spike-and-slab priors (see Castillo, Schmidt-Hieber and van der Vaart (2015), Ishwaran and Rao (2005), Ročková and George (2018), and the references therein) on $\nu_{kg}^{(j)}$. This helps to discard the insignificant promotion effects. Further details are presented in Section 5.4 of the Supplementary Material.

4. Estimation: Algorithmic details and properties. We next describe the algorithm used for estimating the CRON and the SCRON models. There are several statistical challenges associated with the estimation procedure, as the data set is not only large, needing a scalable algorithm, but also the design is incomplete (missing (i, k) pairs; see Table S.1 of the Supplementary Material) and imbalanced. Our proposed algorithm contains several key features, which we explain subsequently.

4.1. *Calibrating the nonlinear effects.* We use a greedy, marginal method for appropriately calibrating the shape of the nonlinear effects of the recency variables. We substitute $b_i = 0$ and $\nu_k^{(j)} = 0$ for all i, k, j in equations (1)–(3). We fit splines separately for the open, click, and purchase stages. We illustrate the procedure for the open stage first. Let $\boldsymbol{R}^o = \{R_{ik}^o : 1 \le i \le n, 1 \le k \le K\}$ be all the different values of days since last opened in the data. Correspondingly, let $\boldsymbol{R}^c$ and $\boldsymbol{R}^p$ denote the vectors of days since last clicked and purchased respectively. Consider placing knots at the deciles of $\boldsymbol{R}^o$, $\boldsymbol{R}^c$ and $\boldsymbol{R}^p$. Fit logistic regression model of (1) (with $b_i, \nu_k^{(1)} = 0$) on a subsample of the training data that contains all the conversions and an equal number of randomly chosen nonconversions. This subsampling is done to correct the imbalance (preponderance of nonconversions) in the data (details are presented in Section 4.2.1 later in a different context). Construct receiver operating characteristic (ROC) curve on the test data based on the above fit. Drop knots iteratively (one at a time) from the center of the data, unless the ROC curve significantly deteriorates. The process is repeated for click and purchase stages with equations (2) and (3). We fix the chosen knots for the three splines and estimate their coefficients along with the promotion effects by the algorithm described in the following section. Note that we have not fitted separate splines for different promotions but have used a global fit for the entire data. Also, we have used intercepts only to measure promotions effects. We have not used intercept coefficients for the spline components (see equation (4)) which would have made them unidentifiable from the promotion effects.

4.2. *Block Metropolis–Hastings with subsampling and transformations.* We first illustrate the log-likelihood involving of the CRON model. Let $B = \{\boldsymbol{b}_i : i = 1, \dots, N\}$ contain all the latent customer preferences and Θ be the set containing all the other parameters used in equations (1)–(4). Thus, Θ includes the promotion effects $\boldsymbol{\nu} = \{\nu_k^{(j)}, k = 1, \dots, K; j = 1, 2, 3\}$, the $\boldsymbol{\beta} = \{\boldsymbol{\beta}^{(j)} : j = 1, 2, 3\}$ as well as the spline coefficients $\mathfrak{m} = \{\boldsymbol{\mu}^{(jl)} : j, l = 1, 2, 3\}$. Denote the log-odds ratio $\ell_{ik}^o := \text{logit}(P_{(\Theta,B)}(O_{ik} = 1))$, the conditional log-odds ratios $\ell_{ik}^c = \text{logit}(P_{(\Theta,B)}(C_{ik} = 1 | O_{ik} = 1))$, and $\ell_{ik}^p = \text{logit}(P_{(\Theta,B)}(P_{ik} = 1 | C_{ik} = 1))$, as given in equations (1)–(3). Let Ω be the set of (i, k) pairs in our training data $\mathcal{D}$, that is, $\Omega = \{(i, k) \in \mathcal{D} : i = 1, \dots, n; k = 1, \dots, K\}$. Then, the log-likelihood $l(\Theta, B | \boldsymbol{o}, \boldsymbol{c}, \boldsymbol{p})$ is

given by

$$\sum_{(i,k)\in\Omega} [o_{ik}\ell_{ik}^o + c_{ik}\ell_{ik}^c + p_{ik}\ell_{ik}^p - \log(1+\exp(\ell_{ik}^o)) - o_{ik}\log(1+\exp(\ell_{ik}^c)) - c_{ik}\log(1+\exp(\ell_{ik}^p))]. \tag{5}$$

On Θ, we use a independent normal prior $g_1 = N(\boldsymbol{\theta_0}, \operatorname{diag}(\boldsymbol{\sigma}_0^2))$. The hyperparameters $\boldsymbol{\theta}_0$, $\boldsymbol{\sigma}_0^2$ are set based on GLM estimates fitted separately for the three stages. On B, we use prior $g_2(B|\Sigma) = \prod_{i=1}^n \phi_3(\boldsymbol{b}_i|\mathbf{0}, \Sigma)$ where $\phi_3(\cdot|\mathbf{0}, \Sigma)$ is a trivariate normal distribution with mean $\mathbf{0}$ and variance Σ. An inverse-Wishart prior g_3 is imposed on Σ. We generate Markov chain Monte Carlo (MCMC) samples from the posterior density whose logarithm is

$$d(\Theta, B) := l(\Theta, B|\boldsymbol{o}, \boldsymbol{c}, \boldsymbol{p}) + \log g_1(\Theta) + \log g_2(B|\Sigma) + \log g_3(\Sigma). \tag{6}$$

4.2.1. *Subsampling and imbalanced data.* An issue with sampling directly from $d(\Theta, B)$ is that, in these applications, there are often much lower numbers of conversions (cases) than nonconversions (controls). Although this marginal imbalance in the data is pervasive in all three stages of the conversion funnel, it is acute in the purchase stage, and $\sum_{(i,k)\in\Omega} p_{ik}/\sum_{(i,k)\in\Omega}(1-p_{ik})$ as well as $\sum_{(i,k)\in\Omega} p_{ik}/\sum_{(i,k)\in\Omega} c_{ik}$ are usually quite small. Traditional frequentist as well as Bayesian approaches for estimating logistic regression models in imbalanced data sets lead to erroneous results (see Johndrow et al. (2019), Owen (2007) and the references therein). A popular fix (Anderson (1972), Chawla, Japkowicz and Kotcz (2004), Prentice and Pyke (1979)) is to conduct *case-control subsampling* (Fithian and Hastie (2014)) so that the ratio of conversions to nonconversions increases in the sample, and imbalance is reduced.

Case-control subsampling is usually conducted by incorporating all the cases and exactly ρ times as many controls for some fixed ρ, such as $\rho = 1, 2, 5$. We consider a similar procedure based on acceptance-rejection sampling. Consider the following nested subsampling scheme. Instead of using Ω for model fitting, we use the following nested subsets $\Omega \supseteq \Omega_o \supseteq \Omega_c \supseteq \Omega_p$ for estimation in the open, click, and purchase stages. The case-to-control ratio in the open stage is $1:\rho_1$. Conditioned on an email being opened, the case-to-control ratio in the click stage is $1:\rho_2$, and conditioned on a clicked email the case-to-control ratio in the purchase stage is $1:\rho_3$. The nested subsampling scheme is:

- For all $(i,k)\in\Omega$, generate independent Bernoulli random variable Z_{ik}^o such that
$$P(Z_{ik}^o = 1) = o_{ik} + \rho_1(1-o_{ik})\sum_{(i,k)\in\Omega} o_{ik}\Big/\Big(|\Omega| - \sum_{(i,k)\in\Omega} o_{ik}\Big),$$
where $|\Omega| = \sum_{1\le i\le N; 1\le k\le K} I\{(i,k)\in\Omega\}$. Define a subsample Ω_o of Ω as $\Omega_o = \{(i,k)\in\Omega : Z_{ij}^o = 1\}$. It contains all the conversions. The expected proportion of conversions to nonconversions in the sample is ρ_1.
- For all $(i,k)\in\Omega_o$, generate independent Bernoulli random variable Z_{ik}^c such that $P(Z_{ik}^c = 1) = c_{ik} + \rho_2(1-c_{ik})\sum_{(i,k)\in\Omega_o} c_{ik}/(|\Omega_o| - \sum_{(i,k)\in\Omega_o} c_{ik})$, where $|\Omega_o|$ denotes the cardinality of Ω_o. Define $\Omega_c = \{(i,k)\in\Omega_o : Z_{ik}^c = 1\}$.
- For all $(i,k)\in\Omega_c$, generate independent Bernoulli random variable Z_{ik}^p such that $P(Z_{ik}^p = 1) = p_{ik} + \rho_3(1-p_{ik})\sum_{(i,k)\in\Omega_c} p_{ik}/(|\Omega_c| - \sum_{(i,k)\in\Omega_c} p_{ik})$. Define $\Omega_p = \{(i,k)\in\Omega_c : Z_{ik}^p = 1\}$.

On the subsampled data the dimension of the B can decrease. Set $B^S = \{b_i : (i,k)\in\Omega_o \text{ for some } k\}$. Then, in the subsampled data, the logarithm of the posterior density from equation (6) reduces to

$$d_S(\Theta, B^S) := l_S(\Theta, B^S) + \log g_1(\Theta) + \log g_2(B^S|\Sigma) + \log g_3(\Sigma), \tag{7}$$

where the $l_S(\Theta, B^S)$ is the log-likelihood, based on the sampled data,

$$\begin{aligned} l_S(\Theta, B^S) = &\sum_{(i,k)\in\Omega_o} o_{ik}\ell_{ik}^o + \sum_{(i,k)\in\Omega_c} c_{ik}\ell_{ik}^c + \sum_{(i,k)\in\Omega_p} p_{ik}\ell_{ik}^p \\ &- |\Omega_o|\log\big(1+\exp(\ell_{ik}^o)\big) - |\Omega_c|\log\big(1+\exp(\ell_{ik}^c)\big) \\ &- |\Omega_p|\log\big(1+\exp(\ell_{ik}^p)\big). \end{aligned} \tag{8}$$

Note that $l_S(\Theta, B^S)$ is not an unbiased estimator of $l(\Theta, B|\boldsymbol{o}, \boldsymbol{c}, \boldsymbol{p})$ from equation (5), and, therefore, we need to adjust the posterior estimates produced by sampling from $d_S(\Theta, B^S)$ instead of $d(\Theta, B)$. This is done at the end of the Metropolis–Hastings algorithm below. Also, note that here we have implemented a uniform down sampling of the controls. Fithian and Hastie (2014) showed that subsampling policies that adjusts the class balance locally in feature space can outperform uniform down-sampling. However, such policies use proximity measures in the feature space and cannot be directly extended to our application, due to the high dimensionality of the feature space and joint modeling across the *three* nested stages.

4.2.2. *Block Metropolis–Hastings with additive transformations* (*BMHT*). We next provide updates for sampling from the posterior distribution. These updates are done in two major blocks. We initialize Θ by $\hat{\Theta}_0$ which is selected by running GLM (with b_is set to 0) separately on the three stages in CRON after the knots locations are chosen, based on the prescription in the previous subsection. We set $\hat{B}_0 = 0$ and present updates for $(\hat{\Theta}_t, \hat{B}_t)$, as iteration step t increases.

Block 1: Update Θ, keeping B fixed. This block is updated sequentially for the three stages of the conversion funnel. Write $\hat{\Theta}$ as $\{\hat{\Theta}^{(j)} : j = 1, 2, 3\}$, consisting of parameters from the open, click, and purchase stages, respectively. Denote $\hat{\Theta}^{(-j)} = \hat{\Theta} \setminus \hat{\Theta}^{(j)}$. Note that, given B, $\hat{\Theta}^{(j)}$ is conditionally independent of $\hat{\Theta}^{(-j)}$ in equation (6). Thus, at iteration t and separately for each j, we update from $\hat{\Theta}_t^{(j)}$ to $\hat{\Theta}_{t+1}^{(j)}$ while $\hat{B}_t$ is fixed. Now, noting that $g_1(\hat{\Theta}) = g_{11}(\hat{\Theta}^{(j)})g_{12}(\hat{\Theta}^{(j)})g_{13}(\hat{\Theta}^{(j)})$ for $j = 1, 2, 3$:

(a) Draw $\hat{\Theta}^{(j)}$ by perturbing $\hat{\Theta}_t^{(j)}$ with Gaussian noise, the variance of which is calibrated by looking at mixing in the chain. Then, calculate

$$r_{t+1}^{(j)} = d_S\big(\hat{\Theta}^{(j)}, \hat{\Theta}_t^{(-j)}, \hat{B}_t^S\big) - d_S\big(\hat{\Theta}_t^{(j)}, \hat{\Theta}_t^{(-j)}, \hat{B}_t^S\big);$$

(b) Generate U from Uniform[0, 1]. If $r_{t+1}^{(j)} \le U$, update $\hat{\Theta}_{t+1}^{(j)} = \hat{\Theta}^{(j)}$; else $\hat{\Theta}_{t+1}^{(j)} = \hat{\Theta}_t^{(j)}$.

Block 2: Update customer-specific preferences B, keeping Θ fixed. As cardinality of B^S is large, convergence of the naive MCMC samplers can take super-linear time (Gao and Owen (2017)). Unless great care is taken to properly scale the proposal distribution, the random walk Metropolis (RWM) algorithm can have poor convergence properties. As B^S is large, an issue in updating all $\boldsymbol{b}_i \in B^S$ simultaneously, using i.i.d. draws from $g_2 \cdot g_3$, is that there is always a positive probability that some coordinate of the multiple random draws will be ill-proposed; in that case the acceptance ratio will be extremely small and with high probability will lead to the rejection of the entire move. To enhance the acceptance rate, we implement the transformation-based Markov chain Monte Carlo (TMCMC) algorithm (Dey and Bhattacharya (2019), Dutta and Bhattacharya (2014)) that simultaneously updates all components in B^S, using appropriate move types, as defined by deterministic transformation of a single draw from $g_2 \cdot g_3$. Generate Σ from g_3 and $\hat{\boldsymbol{b}}$ from $\phi(\cdot|\boldsymbol{0}, \Sigma)$. Consider $\hat{B}_{t+1}^S = \hat{B}_t^S + \rho Z_i \hat{\boldsymbol{b}}$, where Z_is are i.i.d. from Bernoulli(0.5) and $\rho > 0$ is a fixed small number. Generate U from Uniform[0, 1]. Now, if $d_S(\hat{\Theta}_{t+1}, \hat{B}_{t+1}^S) - d_S(\hat{\Theta}_t, \hat{B}_t^S) \le U$, then $\hat{B}_t^S = \hat{B}_{t+1}^S$.

The scale ρ in the TMCMC step is chosen based on the convergence diagnostics, as prescribed in Dey and Bhattacharya (2019). Papaspiliopoulos, Roberts and Zanella (2020) have shown fast convergence of collapsed samplers in two-way cross-classified setups, albeit in a balanced linear model. Geometric ergodicity of TMCMC methods has been established by Dey and Bhattacharya (2016). Thus, we expect our proposed algorithm to converge. We show that the above-mentioned block Metropolis–Hastings with additive TMCMC modifications (BMHT) converges. The proof, presented in the Appendix, follows by showing that BMHT satisfies the conditions in Theorem 2.1 of Dey and Bhattacharya (2016) which are necessary and sufficient conditions for geometric convergence of the TMCMC induced Markov chains.

We need the following assumptions on the design matrix. Let $X_S^{(j)}$ denote the covariate matrix corresponding to $\Theta^{(j)}$ for the subsample $S = (\Omega_o, \Omega_c, \Omega_p)$. For any fixed K, any subsample S of Σ, generated following the scheme in Section 4.2.1, obeys the conditions:

ASSUMPTION 1. The data matrix $X_S^{(j)}$ has full column rank for all $j = 1, 2, 3$.

ASSUMPTION 2. ρ_1, ρ_2, ρ_3 are bounded above by a constant C_0. Also, the imbalance at any stage of the conversion funnel for any promotion does not lead to degeneracy; that is, there exists $\epsilon > 0$ such that

$$\min_{1\le k\le K} \min\left(\sum_i a_{ik} I\{(i,k) \in S\}, \sum_i (1 - a_{ik}) I\{(i,k) \in S\}\right) > \epsilon|\Omega|$$

holds when o_{ik}, c_{ik} and p_{ik} are substituted for a_{ik}.

Assumption 2 implies that all promotions lead to some purchase for subsample S. Note that if there are promotions with no purchases in S, then its corresponding effect is not estimable. These assumptions are very standard in regression setups and can be checked for each subsample. We have the following result on convergence of our proposed algorithm. The proof is provided in Section S.5 of the Supplementary Material.

PROPOSITION 1. *Under Assumptions 1 and 2, the BMHT Markov chain is geometrically ergodic.*

4.2.3. *Adjusting estimates from BMHT for case-control subsampling.* After the chain has converged, the posterior distribution from the burnt-in chain, based on the subsampled data, can be converted to a consistent posterior distribution of the model parameter, based on the original data, by simply conducting only a location shift to the intercepts $\alpha^{(j)}$: update $\hat{\alpha}^{(j)}$ to $\hat{\alpha}^{(j)} - \log \rho_j$ for $j = 1, 2, 3$. Let $\hat{\Theta}^S$ denote the posterior mean (after correction) and $\hat{\Sigma}^S$ be the estimates for the variance components obtained from BMHT, based on subsample S and satsifying Assumptions 1–2. The plug-in estimate of predictive distribution (Geisser (1993)) using $\hat{\Theta}$ and $\hat{\Sigma}$ converges in probability to the true distribution and can be used for out-of-sample predictions (see Section S.6 of the Supplementary Material).

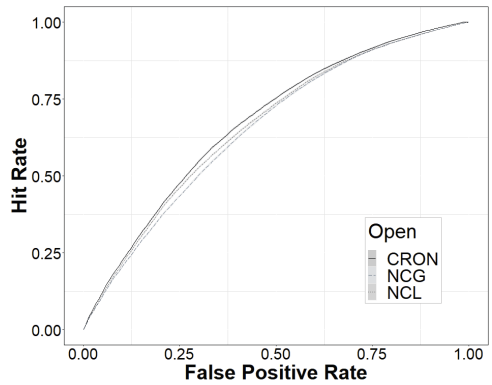
4.3. *Aggregating inference from multiple subsamples.* We repeat the estimation process in Sections 4.2.1 and 4.2.2 for L different subsamples by running Monte Carlo algorithms in separate machines. Akin to Scott et al. (2016), after the chains have burnt-in, we combine the draws from the chains, based on fractional priors, by using weighted averages with each weight being the reciprocal of its marginal posterior variance. A problem with this approach is that the invariant distribution is not preserved across samples. Recent works of Sachs et al. (2020), Sen et al. (2020) develop a novel methodolgy that uses subsampling based unbiased estimates of the gradient of the log-likelihood to arrive at the exact invariant distribution.

However, as we have very large sample sizes here, the distributions from the separate machines will be reasonably close, and we can expect the combined distribution of the TMCMC output from the subsampled chains, each of which are geometrically ergodic, to coverge to the target distribution (see Theorem 2.2 of Alquier et al. (2016)). Combining information from different correlated chains can increase estimation efficiency, as eventually more data is used compared to inference from a separate chain. However, note that if subsampling is used in the open stage, then there are different parameters in B for the individual chains, and we conduct a careful tabulation for aggregating results from the different subsampled TMCMC chains. Details regarding the parallel implementation of the Bayesian estimation methodology is provided in Section S.2.1 of the Supplementary Material.

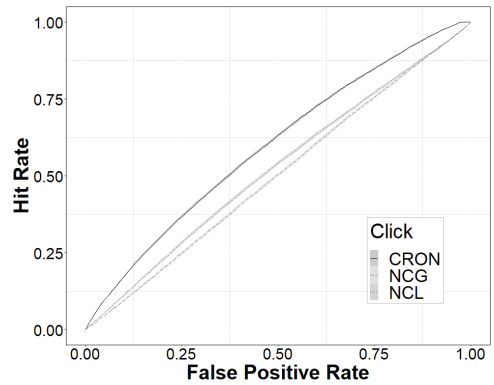
4.4. *Estimating SCRON with spike-and-slab prior.* While estimating SCRON, instead of normal prior we impose $\{v_{kg}^{(j)} : k = 1, \dots, K; g = 1, \dots, G; j = 1, 2, 3\}$ i.i.d. form mixture normal prior $\pi_\eta = (1-\eta)N(0, \tau_1) + \eta N(0, \tau_2)$, where the mixing weight $\eta \sim \text{Beta}(a_1, a_2)$. The hyperparameters τ_1 and τ_2 were chosen as random quantities with τ_1 following inverse gamma distribution with parameters a_σ and b_σ and $\tau_2 = \tau_1/1000$, respectively. The choice of τ_2 was inspired by the procedure prescribed in Malsiner-Walli and Wagner (2011) for tuning spike-and-slab based estimation methodology. For efficient calibration we tried several values of the hyperparameters and chose the optimal setting. Details behind their selection is provided in Section S.4.1 of the Supplementary Material. For open stage, these hyperparameters are taken as $a_\sigma = 2.5$; $b_\sigma = 35$, for click stage $a_\sigma = 2.5$; $b_\sigma = 35$, and for purchase stage $a_\sigma = 2.5$; $b_\sigma = 87.5$. The mixing weight a_1 and a_2 are set to be $(3, 2)$ for open stage, $(2, 2)$ for click stage, and $(7, 3)$ for purchase stage. Promotions parameters with more than 0.5 probability of belonging to the spiked class were deleted from the model, and the resultant model was refitted again using normal priors on the remaining promotion coefficients to produce the final model estimates. Following Bai, Rockova and George (2021), the estimated model inherits decision theoretic guarantees for consistent variable selection.

5. Results and business implications.

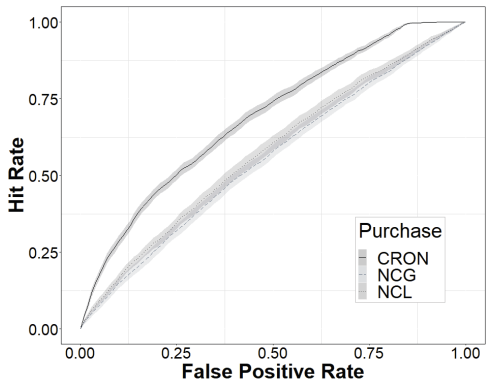
5.1. *ROC curves*: *Test vs. training errors*. We fitted the models NCG, NCL, and CRON, mentioned in the previous section, on the training data. We set $\rho_3 = 4$ and did not use any subsampling in the open and click stages, for those conversion percentages were not very imbalanced in our data. Based on predictions at each stage of the funnel for the test data, Figure 4 shows the out-of-sample receiver operations characteristics (ROC) curves, along with their 95% confidence intervals. The out-of-sample ROC shows the plots of the proportion of correct predictions of conversion (hit rate) vs. the proportion of incorrect predictions of the no-conversions (false positive rate) in the test data. The confidence intervals were constructed by bootstrapping. Table S.14 in the Supplementary Material documents the hit rates from ROC curves, as the false positive proportion is allowed to increase, both for test and training data, wherein we observed that the results in the test data are in accordance with those in the training sample. Based on the areas under the ROC curves (AUROCs) from the test data, we found that NCL improves on the NCG by 2.5% in the open stage, and CRON further improves on NCL by 3.03% (see panel (d) of Figure 4); the improvements are more pronounced in the click and purchase stages, with CRON improving on its nearest competitor by 16.39% and 19.76%, respectively. In Tables S.5, S.6, and S.7 shown in the Supplementary Material, we report posterior means and variances as the coefficient estimates and standard errors for the different models.



ROC of the Open Stage



ROC of the Click Stage



ROC of the Purchase Stage

Stage of the Funnel	NCG		NCL		CRON	
	Mean	SD	Mean	SD	Mean	SD
Open	0.6550	0.0021	0.6717	0.0019	0.6921	0.0017
Click	0.5014	0.0022	0.5278	0.0021	0.6143	0.0019
Purchase	0.5568	0.0066	0.5769	0.0069	0.6909	0.0051

Comparing AUROCs across the funnel

FIG. 4. *Out-of-sample ROC curves along with their* 95% *confidence bands are shown in each subplot for the NCG, NCL, and CRON. Starting from top left, clockwise, we have the ROC curves for open, click, and purchase stages, respectively. The table in the bottom-right panel presents the out-of-sample AUROC estimates and their standard deviations that were calculated by bootstrapping the test data.*

5.2. *Impact of promotions on open rates.* Table S.7, in the Supplementary Material, shows the promotion effect estimates from the CRON model (with respect to the baseline Promo ID 1). We observe that the estimated logodds of opening an email increases around *seven* times in the best promotion, as compared to the worst, provided the other factors are kept constant. The results show that it is extremely important to accurately identify the high-performing promotions. The promotion with the highest conditional effect had "Secret Sale" in the subject of the email; it is well known in the marketing literature that such keywords produce significant hedonic appeal which leads to consumption impulses (Moore and Lee (2012)). Based on estimates, we observe that the top promotions were mostly nonpriced promotions; these had promotion codes for "Cash Refund" or "Buy-One-Get-One Free" in the subject line. The dominance of nonpriced promotions is in accordance with findings that exist in the marketing literature (Park, Park and Schweidel (2018)).

To validate the effectiveness of these promotion-effect estimates $\{\hat{\nu}_k^{(1)} : k = 1, \dots, K\}$ for the open stage, we look at their opening conversion rates in the test data. Based on the descending order of the estimates of promotion-effects from the CRON model (see Table S.7 of the Supplementary Material), we further divide the promotions into three baskets: Top Performing (top eight of the 25 promotions), Medium Performing (promotions ranked between 9–17), and the rest of the promotions (ranked 18–25) as Worse Performing. Table 2 presents their respective conversion rates in the test data. We witness that the rank order of the promo-

tions provided by the estimates (in Table S.7) is well maintained in the conversion efficiency in the test data. We find while the top promotions produce on average 46% conversions, the medium ones yield 41%, while the promotions in the bottom bucket yield 36%.

5.3. *Promotion effects across the conversion funnel*. Using the estimated conditional promotion effects for the click and purchase stages of the CRON model from Table S.7 in the Supplementary Material, we divide the promotions in the click and purchase stages into similar three baskets, following the heuristics mentioned in Section 5.3. On the test data we find that promotions in the top basket produce, on average, 47% click conversions among the opened emails and 12% purchases from the clicked emails. The mid basket produces 34% click conversions and 8% purchases, whereas the bottom basket produces 31% and 5%, respectively. Thus, choosing top promotion basket can lead to 38% more clicks on opened emails and more than double the number of purchases from the clicked emails than the bottom basket. Table 2 presents these detailed conversion results for each promotion on the test data. We observe in this table that the ordering of the estimated promotion effects from CRON (shown in Table S.7 of the Supplementary Material) agree with their performance (rank) in the test data.

TABLE 2

Conversion rates of the promotions ranked in the top, mid, and bottom basket categories from the CRON model. Here, No. of Recipients in the click and purchase stages are the counts of customers who opened and clicked the promotions, respectively

Panel A: Open stage of the conversion funnel

	Open stage				
Promo. efficiency	Promo. no.	No. of recipients	Opened	Open rate (%)	Avg. open rate (%)
Top	13	9980	5244	53	46
Performing	17	5070	2642	52	
Promotions	21	9814	3591	37	
	22	9885	5145	52	
	16	9638	4137	43	
	10	2019	817	40	
	20	4659	1866	40	
	2	9728	4266	44	
Medium	4	9774	4362	45	41
Performing	5	7670	3408	44	
Promotions	25	9836	4493	46	
	1	9665	3507	36	
	11	9885	4051	41	
	9	9770	3566	36	
	6	9763	3910	40	
	8	5036	1962	39	
	18	9791	3439	35	
Worse	14	9909	3515	35	36
Performing	24	6670	2511	38	
Promotions	7	9725	3329	34	
	12	9762	3621	37	
	3	10,000	3805	38	
	23	4842	1775	37	
	15	4968	1660	33	
	19	9758	3222	33	

TABLE 2
(*Continued*)

Panel B: Click and Purchase stages of the conversion funnel										
	Click stage					Purchase stage				
Promo. efficiency	Promo. no.	No. of recipients	No. of clicks	Click rate (%)	Avg. click rate (%)	Promo. no.	No. of recipients	No. of purchases	Pur. rate (%)	Avg. pur. rate (%)
Top	13	5244	3172	60	47	2	2056	327	16	12
Performing	2	4266	2056	48		14	1122	172	15	
Promotions	25	4493	2101	47		21	1238	150	12	
	6	3910	1726	44		8	534	46	09	
	11	4051	1790	44		20	712	76	11	
	10	817	367	45		22	1624	162	10	
	16	4137	1790	43		10	367	23	06	
	9	3566	1431	40		12	1416	132	09	
Medium	12	3621	1416	39	34	17	829	54	07	08
Performing	5	3408	1329	39		6	1726	161	09	
Promotions	20	1866	712	38		1	982	73	07	
	21	3591	1238	34		18	1009	84	08	
	14	3515	1122	32		16	1790	137	08	
	4	4362	1430	33		9	1431	113	08	
	22	5145	1624	32		5	1329	79	06	
	24	2511	762	30		7	857	65	08	
	17	2642	829	31		4	1430	97	07	
Worse	1	3507	982	28	31	11	1790	90	05	05
Performing	18	3439	1009	29		13	3172	177	06	
Promotions	7	3329	857	26		25	2101	90	04	
	8	1962	534	27		24	762	4	01	
	3	3805	1659	44		3	1659	131	08	
	15	1660	535	32		15	535	21	04	
	23	1775	583	33		23	583	39	07	
	19	3222	838	26		19	838	15	02	

Tables 2A and B show that comparative efficiency of a promotion can greatly vary across the conversion funnel. Figure 5 plots the estimates from the CRON model for the different stages of the funnel. For example, *Promo* 13 (with the subject line "Secret Sale") estimates

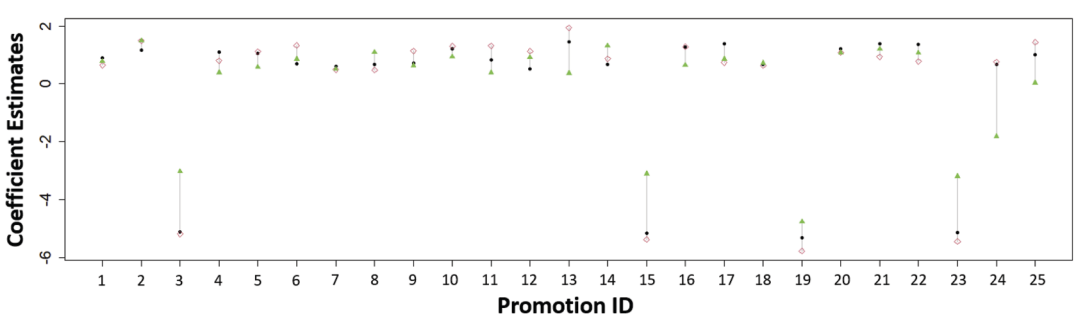
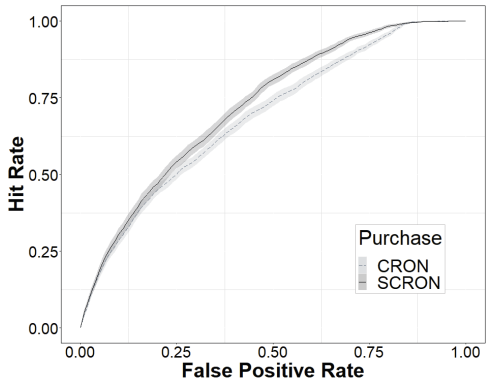


FIG. 5. *Estimates of promotion effects in the CRON model across open* (*circles*), *click* (*paralleleograms*), *and purchase* (*uptriangles*) *stages are plotted. The means of the promotion effects are centered to* 0 *in each stage of the conversion funnel.*



ROC of the Purchase Stage

Stage of the Funnel	CRON		SCRON	
	Mean	SD	Mean	SD
Open	0.6921	0.0017	0.6923	0.0017
Click	0.6143	0.0019	0.6171	0.0019
Purchase	0.6909	0.0051	0.7248	0.0052

AUROCs across the funnel

FIG. 6. *Comparative performance of SCRON and CRON in test data. On left, ROC curve for out-of-sample prediction along with its 95% confidence band is plotted for the purchase stage. The table on the right reports the out-of-sample AUROC estimates as well as their standard deviations from the concerned models.*

are the highest in open and click stages, but its conditional impact in the purchase stage is low (ranked 18 out of 25). Such promotion will be very useful for brand awareness but may not be very effective for purchase conversion. Thus, CRON estimates will be helpful for marketers to determine which promotions to use for different functional purposes.

5.4. *Variation in promotion effects across customer segments.* Based on traditional marketing practice and without using the customer responses to focal promotions in the current data (Gopalakrishnan and Park (2021), Zhang, Kumar and Cosguner (2017)), the customers are binned in $G = 3$ groups: low, mid, and highly active types, based on a customer's engagement history with the firm. Tables S.8–S.10 of the Supplementary Material show the average attributes from these three customer segments. We do not observe much difference in the coverage of the promotions across the three groups. Also, we do not observe severe differences across the means of the covariates pertaining to different promotions that could lead to highly biased estimates of the parameters in the SCRON model. We fit the SCRON model (coefficient estimates are presented in Tables S.11, S.12, S.13 of the Supplementary Material) that allows promotion effects to vary not only within but also between these customer segments. While the performance of SCRON with respect to CRON was similar in the open and click stages, the SCRON performed significantly better in the purchase stage and had 4.91% higher AUROC in the test data, as compared to CRON (as shown in Figure 6).

Targeted marketing aims to design high-performing advertisements and promotions suitably tailored for different customer groups. Figure S.2 of the Supplementary Material shows how the promotion-effect estimates vary across the segments. It also shows that the promotion effects from the high-engagement segment are consistently higher than those from mid and low segments, particularly in the open stage (in the other two stages we have conditional effects based on the open stage). Based on these promotion estimates from SCRON, top-performing promotions in different customer segments can be listed. Marketers can use these lists for targeted marketing. Next, we present an application demonstrating that significant gains can be achieved by targeted marketing strategies based on the SCRON model.

5.5. *Profitability analysis using targeted promotions.* In this marketing campaign, customers received on average around 21 emails in a very short campaign window (see Table S.1 of the Supplementary Material). Marketing research shows that such high frequency of emails can lead to customer fatigue (Statista (2019)). Highly frequent email promotion campaigns

TABLE 3
Conversion Rates of SCRON and CRON based promotion campaign with only five emails

Panel A: Conversion rates in the test data for SCRON based personalized email campaign and CRON based unsegmented email promotions. The last row shows the lift of the former method based on the later

Segments	Open rate (%)	Click rate (%)	Purchase rate (%)
SCRON in High Engagement group	48.25	44.56	12.95
SCRON in Mid Engagement group	45.39	38.94	05.32
SCRON in Low Engagement group	44.55	41.99	03.03
SCRON combined across groups	47.19	42.88	10.17
CRON with no segments	46.77	41.68	07.86
Lift in SCRON vs. CRON	00.90	02.88	29.44

Panel B: Detailed conversion rates of the different promotions chosen by the SCRON model in the three different consumer segments and by the CRON model across all consumers. These conversion rates on aggregation produce the table in panel A

Segment	Promo ID	Count	Open	Click	Purchase	Open rate (%)	Click rate (%)	Purchase rate (%)
SCRON:	13	6104	3345	2107	152	55	63	07
High	17	2872	1539	489	37	54	32	08
Engagement	22	6047	3215	1017	132	53	32	13
	2	5907	2719	1416	314	46	52	22
	21	6009	2181	763	115	36	35	15
SCRON:	21	2714	1001	335	29	37	34	09
Mid	13	2763	1350	763	20	49	57	03
Engagement	22	2736	1384	425	25	51	31	06
	17	1484	739	231	15	50	31	07
	20	1358	544	200	15	40	37	07
SCRON:	21	1091	409	140	06	38	34	04
Low	16	1023	452	201	05	44	44	02
Engagement	13	1113	549	302	05	49	55	02
	20	616	247	100	07	40	40	07
	22	1102	546	182	05	50	33	03
CRON:	13	9980	5244	3172	177	53	60	06
No	17	5070	2642	829	54	52	31	07
Segments	21	9814	3591	1238	150	37	34	12
	22	9885	5145	1624	162	52	32	10
	16	9638	4137	1790	137	43	43	08

often lose relevance and are treated as spam emails. Reducing the number of emails sent to each customers during a campaign is important. We next consider sending only five promotion emails to customers. Based on the promotion-effect estimates from the SCRON model, we construct three different lists of the top five promotions for the low, mid, and highly engaged customer segments. In Table 3A we report their conversion percentages in the test data. We compare the results to the nontargeted marketing policy that uses a list of top five promotions, based on the CRON model, uniformly for all customers. Compared to the unsegmented CRON model, using SCRON we witness marginal improvements in open (1%) and click (3%) stages and significant gains (29%) in the purchase stage (see Table 3B for details).

We next compare the performance of SCRON with state-of-the-art machine learning (ML) methods such as random forests (RF) and XGBoost (XGB). We used the same training data and the same set of variables, as in our SCRON model, to train three unconditional RF models and XGB models. Note that, unlike SCRON, both RF and XGB fit marginal models for the three stages of the conversion funnel. The RF and XGB models are optimized separately for the three stages. Details regarding these procedure are presented in Section S.4.5 of the Supplementary Material. Table S.15 of the Supplementary Material shows the AUROCs, calculated on the test data, for the three different stages of the conversion funnel, based on the RF and XGB models. Comparing Table S.15 of the Supplementary Material with Figure 6, we find that the optimally tuned RF and XGB models are performing better than SCRON in the open stage. However, we find that, as one passes through the subsequent stages of the conversion funnel, namely, click and purchase, the performance of RF and XGB worsens relative to SCRON. We repeat the profitability analysis exercise reported in Table 3 with the RF and XGB models. The results are reported in the Tables S.16 and S.17 of the Supplementary Material. We found that RF and XGB models slightly outperform our proposed joint model in the open stage but are highly suboptimal for the purchase stage. Relative to SCRON, RF predictive performance was 4% better in the open stage and 37% worse in the purchase stage. Similarly, XGB performed 2% better in the open stage but 27% worse than SCRON in the purchase stage. Based on these results, we see that the main advantage of our GLMM based joint modeling approach SCRON is in the purchase stage. RF and XGB perform slightly better or similarly in open and click stages. Due to severe imbalance between cases and controls, the information on the purchase stage is much less than the information in the data regarding the open or click stages. However, though the data strength in the purchase stage is the least, the purchase stage is the most important stage of the funnel, as it is directly linked with revenue. Compared to very powerful nonparametric machine learning methods (RF and XGB trained on similar covariates), the much better performance of SCRON in the purchase stage—the final stage of the funnel with the least data strength—can be attributed to: (a) efficient case-control imbalance correction by SCRON, (b) leveraging the conditional information from the previous stages of the funnel which the unconditional models based on RF and XGB do not use. Also, another advantage of using GLMM based SCRON joint modeling framework is that, unlike RF and XGB, the fitted SCRON model is interpretable and can be readily used by managers to understand efficacy of different promotion emails across various market segments.

6. Discussion. We developed a large-scale, cross-classified, nested joint model for modeling customer responses to opening, clicking, and purchasing from promotion emails, thereby estimating the efficacy of these marketing interventions for customers with varying characteristics and engagement history. We estimate the parameters in the joint model by using a block Metropolis–Hastings algorithm that uses nested subsampling to tackle the severe imbalance between conversions and no conversions. The proposed estimation methodology being based on standard case-control sampling can be inconsistent under model misspecification (Fithian and Hastie (2014)). Further introspection is needed to understand the estimation accuracy of the proposed algorithm under misspecification.

As future work, it will be useful to explore whether our estimation algorithm for the logistic-regression-based joint modeling framework in cross-classified models can be made faster by using data augmentation with Polya-Gamma latent variables, as in Polson, Scott and Windle (2013). It will be particularly interesting to explore the recent approach in Sen et al. (2020) that uses subsample based unbiased estimator of the gradient of the log-likelihood and has the advantage of working with the exact invariant distribution.

We proved geometric ergodicity of our proposed MCMC-based estimation algorithm. Following Bissiri, Holmes and Walker (2016), Kleijn (2012), it will be interesting to introspect

posterior concentration of the estimates, particularly under misspecification. Our joint modeling framework do not involve interactions between successive promotion emails, as we assumed that the impacts of these emails follows a memoryless process, as theorized in Zhang, Kumar and Cosguner (2017). It will be interesting to estimate the impact of a chain/sequence of promotion emails in future data sets where such sequences, which can be parameterized in our framework by higher order interactions among promotion emails, are estimable.

Acknowledgments. We are grateful to Professor Edward George for numerous stimulating discussions and several helpful comments. We thank the Editor, the Associate Editor, and three referees for constructive suggestions that greatly improved the paper.

Funding. The research here was partly supported by NSF Grant DMS-1811866.

SUPPLEMENTARY MATERIAL

Supplementary Material to "Estimating promotion effects in email marketing using a large-scale cross-classified Bayesian joint model for nested imbalanced data" (DOI: 10.1214/22-AOAS1638SUPP; .pdf). The *Supplementary Material* (Mukhopadhyay, Kar and Mukherjee (2023)) contains further details regarding the data set, implementation of the estimation methodology, the fitted models and their operational characteristics. The proof of Proposition 1 is also presented in the *Supplementary Material*.

REFERENCES

ALDIGHIERI, R. (2019). Marketer email tracker 2019. https://dma.org.uk/uploads/misc/marketers-email-tracker-2019.pdf.

ALQUIER, P., FRIEL, N., EVERITT, R. and BOLAND, A. (2016). Noisy Monte Carlo: Convergence of Markov chains with approximate transition kernels. *Stat. Comput.* **26** 29–47. MR3439357 https://doi.org/10.1007/s11222-014-9521-x

ANDERSON, J. A. (1972). Separate sample logistic discrimination. *Biometrika* **59** 19–35. MR0345332 https://doi.org/10.1093/biomet/59.1.19

BAI, R., ROCKOVA, V. and GEORGE, E. I. (2021). Spike-and-slab meets LASSO: A review of the spike-and-slab LASSO. In *Handbook of Bayesian Variable Selection* 81–108.

BISSIRI, P. G., HOLMES, C. C. and WALKER, S. G. (2016). A general framework for updating belief distributions. *J. R. Stat. Soc. Ser. B. Stat. Methodol.* **78** 1103–1130. MR3557191 https://doi.org/10.1111/rssb.12158

BULT, J. R. and WANSBEEK, T. (1995). Optimal selection for direct mail. *Mark. Sci.* **14** 378–394.

CASTILLO, I., SCHMIDT-HIEBER, J. and VAN DER VAART, A. (2015). Bayesian linear regression with sparse priors. *Ann. Statist.* **43** 1986–2018. MR3375874 https://doi.org/10.1214/15-AOS1334

CHAWLA, N. V., JAPKOWICZ, N. and KOTCZ, A. (2004). Special issue on learning from imbalanced data sets. *ACM SIGKDD Explor. Newsl.* **6** 1–6.

DEY, K. K. and BHATTACHARYA, S. (2019). A brief review of optimal scaling of the main MCMC approaches and optimal scaling of additive TMCMC under non-regular cases. *Braz. J. Probab. Stat.* **33** 222–266. MR3919022 https://doi.org/10.1214/17-bjps386

DEY, K. KR. and BHATTACHARYA, S. (2016). On geometric ergodicity of additive and multiplicative transformation-based Markov Chain Monte Carlo in high dimensions. *Braz. J. Probab. Stat.* **30** 570–613. MR3582391 https://doi.org/10.1214/15-BJPS295

DUTTA, S. and BHATTACHARYA, S. (2014). Markov chain Monte Carlo based on deterministic transformations. *Stat. Methodol.* **16** 100–116. MR3110892 https://doi.org/10.1016/j.stamet.2013.08.006

FADER, P. S., HARDIE, B. G. and LEE, K. L. (2005). Counting your customers the easy way: An alternative to the Pareto/NBD model. *Mark. Sci.* **24** 275–284.

FITHIAN, W. and HASTIE, T. (2014). Local case-control sampling: Efficient subsampling in imbalanced data sets. *Ann. Statist.* **42** 1693–1724. MR3257627 https://doi.org/10.1214/14-AOS1220

FORTE, D. (2019). Consumers want more personalized emails, but retailers aren't delivering. Available at https://multichannelmerchant.com/marketing/consumers-want-personalized-emails-retailers-arent-delivering/.

GAO, K. (2017). *Scalable Estimation and Inference for Massive Linear Mixed Models with Crossed Random Effects*. ProQuest LLC, Ann Arbor, MI. Thesis (Ph.D.)–Stanford University. MR4257222

GAO, K. and OWEN, A. (2017). Efficient moment calculations for variance components in large unbalanced crossed random effects models. *Electron. J. Stat.* **11** 1235–1296. MR3635913 https://doi.org/10.1214/17-EJS1236

GEISSER, S. (1993). *Predictive Inference*: *An Introduction*. *Monographs on Statistics and Applied Probability* **55**. CRC Press, New York. MR1252174 https://doi.org/10.1007/978-1-4899-4467-2

GOPALAKRISHNAN, A. and PARK, Y.-H. (2021). The impact of coupons on the visit-to-purchase funnel: Theory and empirical evidence. *Mark. Sci.*.

HUANG, Z. and GELMAN, A. (2005). Sampling for Bayesian computation with large datasets. Available at SSRN 1010107.

ISHWARAN, H. and RAO, J. S. (2005). Spike and slab variable selection: Frequentist and Bayesian strategies. *Ann. Statist.* **33** 730–773. MR2163158 https://doi.org/10.1214/009053604000001147

JAMES, G., WITTEN, D., HASTIE, T. and TIBSHIRANI, R. (2013). *An Introduction to Statistical Learning*. *Springer Texts in Statistics* **103**. Springer, New York. With applications in R. MR3100153 https://doi.org/10.1007/978-1-4614-7138-7

JOHNDROW, J. E., SMITH, A., PILLAI, N. and DUNSON, D. B. (2019). MCMC for imbalanced categorical data. *J. Amer. Statist. Assoc.* **114** 1394–1403. MR4011787 https://doi.org/10.1080/01621459.2018.1505626

KAMAKURA, W. A. and KANG, W. (2007). Chain-wide and store-level analysis for cross-category management. *J. Retail.* **83** 159–170.

KAPNER, S. (2017). Retailers' emails are misfires for many holiday shoppers.

KLEIJN, B. J. K. and VAN DER VAART, A. W. (2012). The Bernstein-von-Mises theorem under misspecification. *Electron. J. Stat.* **6** 354–381.

KUHN, M. and JOHNSON, K. (2013). *Applied Predictive Modeling*. Springer, New York. MR3099297 https://doi.org/10.1007/978-1-4614-6849-3

LEE (2019). Digital vs traditional media—where should you invest your marketing dollars? Available at https://hookdpromotions.com/digital-vs-traditional-media-where-should-you-invest-your-marketing-dollars/.

LEONE, C. (2020). How much should you budget for marketing in 2020. Available at https://www.webstrategiesinc.com/blog/how-much-budget-for-misc-marketing-in-2014.

MALSINER-WALLI, G. and WAGNER, H. (2011). Comparing spike and slab priors for Bayesian variable selection. *Aust. J. Stat.*

MCCAFFREY, D. F., GRIFFIN, B. A., ALMIRALL, D., SLAUGHTER, M. E., RAMCHAND, R. and BURGETTE, L. F. (2013). A tutorial on propensity score estimation for multiple treatments using generalized boosted models. *Stat. Med.* **32** 3388–3414. MR3074364 https://doi.org/10.1002/sim.5753

MCCULLOCH, C. (2008). Joint modelling of mixed outcome types using latent variables. *Stat. Methods Med. Res.* **17** 53–73. MR2420190 https://doi.org/10.1177/0962280207081240

MOORE, D. J. and LEE, S. P. (2012). How advertising influences consumption impulses. *J. Advert.* **41** 107–120.

MUKHOPADHYAY, S., KAR, W. and MUKHERJEE, G. (2023). Supplement to "Estimating promotion effects in email marketing using a large-scale cross-classified Bayesian joint model for nested imbalanced data." https://doi.org/10.1214/22-AOAS1638SUPP

OSINGA, E. C., LEEFLANG, P. S. and WIERINGA, J. E. (2010). Early marketing matters: A time-varying parameter approach to persistence modeling. *J. Mark. Res.* **47** 173–185.

OWEN, A. B. (2007). Infinitely imbalanced logistic regression. *J. Mach. Learn. Res.* **8** 761–773. MR2320678

PAPASPILIOPOULOS, O., ROBERTS, G. O. and ZANELLA, G. (2020). Scalable inference for crossed random effects models. *Biometrika* **107** 25–40. MR4064138 https://doi.org/10.1093/biomet/asz058

PARK, C. H., PARK, Y.-H. and SCHWEIDEL, D. A. (2018). The effects of mobile promotions on customer purchase dynamics. *Int. J. Res. Mark.* **35** 453–470.

PERRIN, N. (2019). Email marketing 2019: Still a leading touchpoint for marketers and consumers alike. Available at: https://content-na1.emarketer.com/email-marketing-2019.

POLSON, N. G., SCOTT, J. G. and WINDLE, J. (2013). Bayesian inference for logistic models using Pólya-Gamma latent variables. *J. Amer. Statist. Assoc.* **108** 1339–1349. MR3174712 https://doi.org/10.1080/01621459.2013.829001

PRENTICE, R. L. and PYKE, R. (1979). Logistic disease incidence models and case-control studies. *Biometrika* **66** 403–411. MR0556730 https://doi.org/10.1093/biomet/66.3.403

RIZOPOULOS, D. (2012). *Joint Models for Longitudinal and Time-to-Event Data*: *With Applications in R*. CRC Press, Boca Raton.

RIZOPOULOS, D. and GHOSH, P. (2011). A Bayesian semiparametric multivariate joint model for multiple longitudinal outcomes and a time-to-event. *Stat. Med.* **30** 1366–1380. MR2828959 https://doi.org/10.1002/sim.4205

RIZOPOULOS, D. and LESAFFRE, E. (2014). Introduction to the special issue on joint modelling techniques. *Stat. Methods Med. Res.* **23** 3–10. MR3190684 https://doi.org/10.1177/0962280212445800

ROČKOVÁ, V. and GEORGE, E. I. (2018). The spike-and-slab LASSO. *J. Amer. Statist. Assoc.* **113** 431–444. MR3803476 https://doi.org/10.1080/01621459.2016.1260469

ROSSI, P. E., MCCULLOCH, R. E. and ALLENBY, G. M. (1996). The value of purchase history data in target marketing. *Mark. Sci.* **15** 321–340.

RUPPERT, D. (2002). Selecting the number of knots for penalized splines. *J. Comput. Graph. Statist.* **11** 735–757. MR1944261 https://doi.org/10.1198/106186002321018768

SACHS, M., SEN, D., LU, J. and DUNSON, D. (2020). Posterior computation with the gibbs zig-zag sampler. Preprint. Available at arXiv:2004.04254.

SAHNI, N. S., WHEELER, S. C. and CHINTAGUNTA, P. (2018). Personalization in email marketing: The role of noninformative advertising content. *Mark. Sci.* **37** 236–258.

SAHNI, N. S., ZOU, D. and CHINTAGUNTA, P. K. (2017). Do targeted discount offers serve as advertising? Evidence from 70 field experiments. *Manage. Sci.* **63** 2688–2705.

SCOTT, S. L., BLOCKER, A. W., BONASSI, F. V., CHIPMAN, H. A., GEORGE, E. I. and MCCULLOCH, R. E. (2016). Bayes and big data: The consensus Monte Carlo algorithm. *Int. J. Manag. Sci. Eng. Manag.* **11** 78–88.

SEETHARAMAN, P. B. and CHINTAGUNTA, P. K. (2003). The proportional hazard model for purchase timing: A comparison of alternative specifications. *J. Bus. Econom. Statist.* **21** 368–382. MR1982528 https://doi.org/10.1198/073500103288619025

SEN, D., SACHS, M., LU, J. and DUNSON, D. B. (2020). Efficient posterior sampling for high-dimensional imbalanced logistic regression. *Biometrika* **107** 1005–1012. MR4186502 https://doi.org/10.1093/biomet/asaa035

STATISTA (2019). Dossier on e-mail marketing in the U.S. and worldwide.

WANG, H., ZHU, R. and MA, P. (2018). Optimal subsampling for large sample logistic regression. *J. Amer. Statist. Assoc.* **113** 829–844. MR3832230 https://doi.org/10.1080/01621459.2017.1292914

WU, J., LI, K. J. and LIU, J. S. (2018). Bayesian inference for assessing effects of email marketing campaigns. *J. Bus. Econom. Statist.* **36** 253–266. MR3790212 https://doi.org/10.1080/07350015.2016.1141096

ZHANG, X., KUMAR, V. and COSGUNER, K. (2017). Dynamically managing a profitable email marketing program. *J. Mark. Res.* **54** 851–866.

ZHANG, X., ZHOU, Y., MA, Y., CHEN, B.-C., ZHANG, L. and AGARWAL, D. (2016). Glmix: Generalized linear mixed models for large-scale response prediction. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 363–372.

Supplement to "Estimating Promotion Effects in Email Marketing using a Large-scale Cross-Classified Bayesian Joint Model for Nested Imbalanced Data"

Sabyasachi Mukhopadhyay, Indian Institute of Management
and
Wreetabrata Kar, Purdue University
and
Gourab Mukherjee, University of Southern California

April 26, 2022

Abstract

Section 1 of the supplement contains further information on the dataset. Tables showing estimated coefficients and standard errors for the NCG, NCL, and CRON models are presented in Section 2. Section 3 presents details regarding implementation of the Bayesian estimation algorithm. Section 4 contains additional information on hyper-parameter calibration, fit, and validation ROC of the SCRON model. Results on test error and profitability analysis of unconditional models based on random forest and XGBoost are presented in Section 4. The proofs of the results stated in the main paper are presented in Sections 5 and 6.

S.1 Further information regarding the data set

The data for this study was collected from an online apparel retailer. Table S.1 below show the distribution of these 25 promotion emails across the 77986 customers studied in this data set.

Table S.1: Distribution of the number of promotion emails received by customers during the concerned campaign.

No of Promotions	**No.of Recipients**
4	2
5	4
6	18
7	18
8	25
9	40
10	66
11	191
12	392
13	668
14	713
15	450
16	363
17	619
18	2196
19	7034
20	15127
21	20518
22	17631
23	9160
24	2461
25	290
Total	77986

In Figure S.1, we provide the above distribution but in terms of percentage of customers

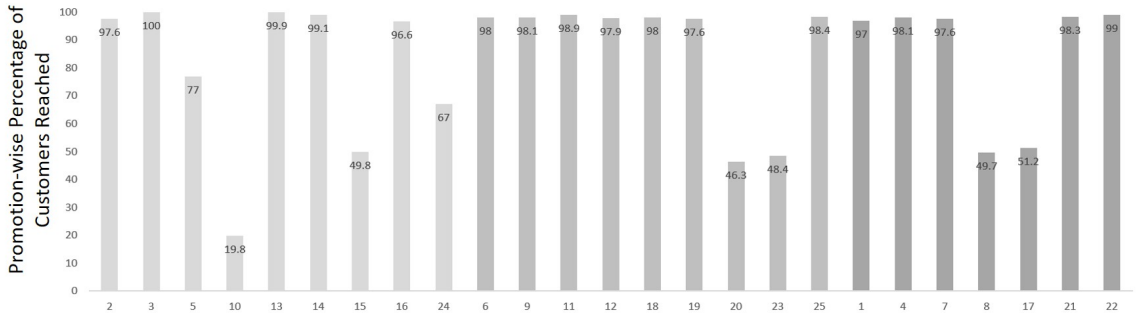


Figure S.1: Percentage of total customers reached by different promotion emails. The promotion number are in congruence with those in Table S.2. In light gray, black and dark gray respectively are 9 priced (in lightgray), 7 non-priced (in darkgray) and 9 combination (in gray) promotions.

reached by each email segregated by type of promotion (price, non-price, and combo). In this figure, *reach* is defined as the number of customers receiving an email divided by the total number of customers in our dataset.

For each of these promotion emails, Table S.2 below shows the summary statistics for the unconditional response rates of each stage of the conversion funnel (open, click, purchase), as well as for the covariates corresponding to the recency, frequency, and monetary variables.

We did not know of any targeting being used in the promotion emails sent out during the campaign period. Some of the promotion emails such as Promotion # 3 (See Fig S.1) were sent to all the customers and barring one promotion, most promotions had a coverage of at least 45% of the consumers. However, based on our discussion with the data vendor, we know for sure that the set of covariates in this paper includes all the variables that the firm would use to make marketing intervention decisions (if any). Nonetheless, to test similarity of customer characteristics across promotions, we compare the balance in the covariates between different promotions following the methodology developed in McCaffrey et al. [2013]. As in our case, we do not have customers who did not receive any promotions at all, we consider Promotion # 1 as the baseline or control promotion. For customer $i = 1, \ldots, N$, let $I_k(i)$ be an binary variable indicating whether a customer i got promotion k and $\widehat{e}_k(i)$ be his estimated propensity-score of receiving promotion k based on a logistic regression model. The standardized bias or SB of the jth covariate for the kth promotion

is defined as follows:

$$\mathrm{SB}_{jk} = |\bar{X}_{jk} - \bar{X}_{j1}|/\hat{\sigma}_j \ , \tag{1}$$

where, $\bar{X}_{jk}$ is the propensity-score weighted mean of the covariate j for promotion k:

$$\bar{X}_{jk} = \left(\sum_{i=1}^{n} I_k(i) x_{ij} / \widehat{e}_k(i) \right) \Big/ \left(\sum_{i=1}^{n} I_k(i) / \widehat{e}_k(i) \right), \text{ for } k = 1, \ldots, K,$$

and $\widehat{\sigma}_j$ is the standard deviation of covariate j over all customers.

Table S.3 displays the propensity-score weighted means $\bar{X}_{jk}$s for promotion $k = 2, \ldots, 25$. We calculate the SB_{jk} value and use asterisk(s) to indicate the degree to which the balance is affected. McCaffrey et al. [2013] note that standardized mean differences of less than 0.20 are considered small, 0.40 are considered moderate, and 0.60 are considered large [Cohen, 2013]. In this exercise, we find excellent balance between promotions, with 95% of the comparisons showing SB score of less than 0.2, and the rest 5%, which is only 9 out of 192 total comparisons, showing moderate balance. In absence of any knowledge of targeting, the strong overlaps in the covariate characteristics of the customers across the different promotions as seen in this standardized bias calculation provide support for our missing at random assumption. Throughout this paper, we assume that the unobserved customer-promotions (i, k) pairs in our data are *missing at random.*

Validation data. The dataset used in this paper contained 25 promotion emails and 77986 customers. We randomly selected 10000 customers as validation data and used responses from the remaining customers for training. As the training data set is huge and the Bayesian estimation methodology employed for estimating the models is extremely computation extensive, it was difficult to conduct a full scale cross-validation which would need estimating these models several times on different subsets of the data. We felt that a random subset of 10000 customers (which approximately contains 2×10^6 responses to promotion emails) will not only be a significantly large validation sample but also leave a reasonably large training data. The validation sample size of 10000 is an arbitrary choice and the only reason behind this choice was to have a validation sample of considerable size and a large training data. Having such large validation and training data sets make us less immune to local idiosyncrasies in the customer pool.

Table S.2: Aggregate statistics averaged across recipients for different promotion emails used in the campaign

Promotions			**Unconditional Responses**			**Days: Last Open**		**Days: Last Clicked**		**Days: Last Purchased**		**Pur. Freq.**		**Avg. Retail**		**Avg. Web**	
ID No	**Type**	**Count**	**Open Rate**	**Click Rate**	**Purchase Rate**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
1	N	75661	0.3589	0.1031	0.0084	9.73	20.41	27.34	91.31	93.38	102.58	7.69	8.63	13.96	30.36	52.08	33.82
2	P	76103	0.4343	0.2113	0.0326	9.35	20.31	26.66	90.06	92.53	101.49	7.70	8.64	13.96	30.34	52.07	33.81
3	P	77986	0.3828	0.1623	0.0132	0.81	8.20	3.67	35.80	68.56	89.06	7.83	8.66	13.92	30.26	52.30	33.50
4	N	76492	0.4451	0.1449	0.0084	8.45	19.25	23.55	84.52	87.61	88.99	7.69	8.63	13.96	30.34	52.04	33.81
5	P	60048	0.4448	0.1742	0.0119	6.56	15.14	17.19	62.24	85.82	90.67	7.79	8.64	14.04	30.11	52.30	33.75
6	C	76398	0.3971	0.1770	0.0157	5.80	14.11	14.46	53.80	84.26	90.84	7.79	8.66	14.02	30.36	52.27	33.70
7	N	76136	0.3404	0.0881	0.0057	6.98	16.04	18.25	66.89	86.30	90.29	7.78	8.64	14.01	30.35	52.28	33.70
8	N	38782	0.3843	0.1009	0.0105	7.35	17.74	20.61	77.11	88.83	89.60	7.70	8.71	13.81	30.08	52.28	34.07
9	C	76470	0.3666	0.1475	0.0103	8.12	18.39	22.71	79.84	87.47	89.20	7.69	8.63	13.96	30.34	52.04	33.81
10	P	15432	0.4128	0.1812	0.0169	8.70	18.67	21.84	66.88	90.81	90.11	7.68	8.54	14.00	30.77	52.17	32.73
11	C	77140	0.4064	0.1783	0.0100	5.05	9.81	10.15	37.07	79.44	87.85	7.87	8.68	13.93	30.27	52.34	33.43
12	C	76385	0.3685	0.1472	0.0136	5.49	13.51	13.43	52.29	85.25	90.83	7.80	8.66	14.02	30.36	52.27	33.70
13	P	77874	0.5260	0.3129	0.0177	5.60	12.18	12.21	46.89	83.68	90.50	7.83	8.65	13.92	30.26	52.32	33.50
14	P	77323	0.3555	0.1169	0.0159	5.59	11.82	12.21	45.54	83.23	90.60	7.83	8.66	13.92	30.27	52.32	33.49
15	P	38808	0.3326	0.1066	0.0045	0.85	5.80	2.81	17.94	70.61	87.68	8.12	8.80	13.99	30.39	52.73	33.54
16	P	75321	0.4220	0.1780	0.0134	6.06	9.64	11.02	36.40	80.79	87.64	7.95	8.66	13.99	30.36	52.50	33.18
17	N	39951	0.5214	0.1573	0.0126	5.04	7.64	9.41	24.13	90.35	96.80	6.08	6.50	12.31	28.41	50.77	32.94
18	C	76403	0.3508	0.0993	0.0077	5.26	9.18	9.74	32.20	78.28	87.30	7.99	8.72	13.98	30.33	52.47	33.14
19	C	76145	0.3293	0.0883	0.0013	0.31	1.65	1.32	3.30	65.46	90.01	8.16	8.80	13.97	30.16	52.32	32.78
20	C	36076	0.3927	0.1447	0.0150	5.83	6.10	10.17	13.77	87.01	96.07	6.72	7.79	12.12	28.30	51.62	33.63
21	N	76633	0.3696	0.1226	0.0145	6.62	5.74	10.81	11.26	79.85	91.22	8.06	8.75	13.98	30.26	52.43	33.10
22	N	77177	0.5212	0.1621	0.0176	4.90	7.11	8.96	21.06	80.88	90.96	7.97	8.71	13.97	30.30	52.42	33.21
23	C	37755	0.3575	0.1119	0.0073	0.62	2.60	2.64	5.06	65.30	86.60	8.30	8.82	13.98	29.82	52.45	32.55
24	P	52242	0.3780	0.1136	0.0007	4.56	6.46	8.94	13.19	82.67	92.16	6.89	7.19	14.93	31.40	53.78	34.12
25	C	76761	0.4546	0.2103	0.0087	4.54	6.10	8.70	12.15	79.57	91.13	8.05	8.74	13.98	30.25	52.41	33.07

Table S.3: Inverse propensity-adjusted average of covariates for each promotion email (barring the base email of Promo ID 1). Each cell in the table, barring column one, shows the customer-specific covariates of the promotions weighted by the inverse probability of getting the promotion. Based on these inverse-propensity adjusted covariate averages, we compute the standardized bias (SB_{kj}) of the jth covariate of the kth promotion compared to the base promotion email (following Eq 1). Cells marked with: an * denote covariates for which SB_{kj} is greater than 0.20; ** refers to SB_{tk} is greater than 0.40; *** SB_{kj} is greater than 0.60.

Promotion ID No	Ave. Retail Spend	Ave. Web Spend	Pur. Freq.	Age	Income (in '000s)	Days Since Opened	Days Since Clicked	Days Since Purchased
2	14.01	52.02	7.68	49.85	69.14	9.53	26.89	92.89
3	14.13	52.49	7.80	49.88	69.93	2.17**	8.69	73.86
4	14.02	52.02	7.68	49.84	68.12	9.24	25.73	90.19
5	14.15	52.07	7.62	49.85	69.09	12.61**	36.64	92.27
6	14.38	52.13	7.58	49.84	69.08	12.30**	34.54	91.35
7	14.09	52.11	7.68	49.84	69.09	10.36	33.01	90.65
8	13.96	52.07	7.64	49.84	69.07	11.39	31.22	92.68
9	14.02	52.02	7.67	49.84	69.12	9.19	25.60	90.20
10	14.06	52.07	7.66	49.26	72.12	9.79	30.24	92.39
11	14.21	52.05	7.55	49.91	69.21	10.33	23.54	91.33
12	14.42	52.31	7.56	49.84	69.09	12.58**	32.11	93.08
13	14.23	52.12	7.61	49.90	69.25	12.07**	25.52	93.09
14	14.28	52.02	7.62	49.90	69.25	11.65**	25.29	92.85
15	14.21	52.91	7.63	49.58	68.52	3.02**	5.60	79.06
16	14.04	52.15	7.65	49.90	69.21	9.82	22.94	91.76
17	13.76	51.18	7.74	49.96	69.30	6.76	14.65	90.20
18	14.12	51.96	7.67	49.90	69.22	9.54	21.67	89.56
19	13.42	54.09	7.77	49.91	68.71	2.92**	2.50	66.74
20	13.89	51.65	8.00	49.82	69.21	6.41	12.58	87.15
21	14.10	52.14	7.89	49.90	69.17	7.52	13.66	84.47
22	14.32	52.07	7.78	49.91	68.83	6.48	14.16	86.78
23	14.61	51.75	7.94	49.84	69.20	1.82**	4.37	72.67
24	14.69	52.73	7.40	49.91	69.21	6.29	12.72	87.09
25	14.33	52.11	7.86	49.91	69.18	5.96	12.13	84.68

S.2 Further Details on the Estimation Procedure

Here, we have randomly subsampled the non-responses in the purchase stage to tackle imbalance. ρ_3 is set to the maximum degree of imbalance we allowed in the sub-samples. To mitigate biases in coefficient estimates from logistic regression model typically the proportion of cases in case-control subsampling is not lowered below 20% in the sub-samples which translates to the upper bound of 4 on ρ_3. As random subsampling was done on the non-responses and we have a lot of observations (corresponding to non-responses) for each promotions even after sub-sampling, the distribution of covariates U^T corresponding to the fixed effects as well as the covariates in the splines would not change much due to subsampling. In Section 2 of Fithian and Hastie [2014], the authors show that for case-control sampling the coefficients in logistic regression can be corrected by just adjusting the intercepts. We have incorporated this correction in our estimation methodology. However, though equations (8)–(10) in Fithian and Hastie [2014] show that the estimates proposed in Section 4.2.3 of the main paper are consistent, the proposed estimates can be inconsistent under model misspecification. As such section 2.2 of Fithian and Hastie [2014] shows that in a misspecified logistic regression model with a single binary response, the estimates proposed in Section 4.2.3 of our main paper can be inconsistent. With three levels of nested subsampling, such model misspecification inconsistency can be quite pronounced in the joint modeling of open, click and purchase stages. In Section 5 of the main paper, based on a test set we show that the proposed algorithm with case-control subsampling implemented only in the purchase stage provides superior out-of-sample operational characteristics compared to state-of-the-art statistical methods. However, further introspection and theoretical study will be needed to understand the estimation error of the proposed algorithm under misspecification.

The splines are fitted on the recency variables only, viz., Days since Last open, click, and purchase. We have not fitted separate splines for different promotions but have fitted a spline on the entire data for each of the recency variables. Intercepts are used to measure promotions effects. For finding the knots of splines we have used used marginal logistic regression models for the open, click, and purchase stages with the promotion as well as customer specific effects set at 0. Knots were placed at the deciles of the recency variables and were then selected based on the fit statistics. With these knots, the spline coefficients were estimated along with the promotion effects in the Block Metropolis-Hastings algorithm. The promotion and the customer specific effects explain the residuals that can be

explained by the splines and the other fixed effects in $\boldsymbol{\beta}^{(j)}$s. Unless there is severe contrast among the customer profiles receiving different promotions (which is assumed not to exist in our case as explained in Section 2 of the main paper), the spline coefficients (as there are no intercepts) would not influence the estimates of promotion effects.

S.2.1 Implementation Details

The implementation details for the CRON and SCRON models are presented in `https://github.com/gmukherjee/emarketing/blob/master/README.md`; For the results in Section 5 we have repeated the estimation process in Sections 4.2.1 and 4.2.2 for $L = 10$ different subsamples by running Monte Carlo algorithms in parallel. The codes for Bayesian estimation of all the models are run separately in a server equipped with 192 GB of RAM and 56 *Intel(R) Xeon(R) Gold 6130 CPU*s of 2.10 GHz speed. We considered 16000 iterations of the MCMC algorithm and used 6000 iterations for burn in. The entire run times (test + training) of the codes (using R software version 4.0.2) for the different models are listed below:

Table S.4: Run times of different models with 10 parallel MCMC sub-chains

Model	Run time
NCG	1 day 21 hours
NCL	2 days 2 hours
CRON	2 days 13 hours
SCRON	2 days 20 hours

The stabilized posterior draws from combined BMHT runs across different subsamples of the data were stored to compute the reported ROC curves and AUROCs in Figures 4 and 6. ROC curves and AUROCs were computed for each of those draws and then averaged over to arrive at the reported curves and AUROCs estimates. Based on these draws, the posterior mean and variances were computed for the coefficients in the different models and were reported. The 95% CI for the ROC curves and the standard deviation of the AUROCs were computed on the test data by bootstrapping and using the posterior means as coefficients of the fitted model.

S.3 Coefficient Estimates, SE of the fitted NCG, NCL, and CRON models

Table S.5: Stage-wise model estimates of NCG

Coefficients	Open stage		Click stage		Purchase stage	
	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	-0.8818	0.0014	-1.7628	0.0033	-4.4131	0.0053
Ave. Retail Spend	0.0008	0.0001	0.0007	0.0002	0.0031	0.0004
Ave. Web Spend	-0.0003	0.0001	0.0017	0.0004	0.0144	0.0007
Pur. Freq.	0.0041	0.0003	0.0161	0.0012	0.0611	0.0027
Age	0.0865	0.0019	0.1038	0.0171	0.1676	0.0284
Income: 15K - 25K	-0.0024	0.0008	0.0006	0.0016	-0.0122	0.0018
Income: 25K - 35K	0.0131	0.0014	-0.0281	0.0012	-0.1038	0.0018
Income: 35K - 50K	0.0239	0.0028	-0.0342	0.0016	-0.0308	0.0014
Income: 50K - 75K	0.0307	0.0013	-0.044	0.0022	-0.0723	0.002
Income: 75K - 100K	0.0552	0.0010	-0.0664	0.0011	-0.1107	0.0015
Income: 100K - 120K	0.0728	0.001	-0.0686	0.0017	-0.1203	0.0029
Income: 120K - 149K	0.0893	0.0009	-0.0689	0.0006	0.019	0.0012
Income: 150K plus	0.1156	0.0018	-0.0809	0.0028	0.0832	0.0013
Days Since Last Opened	0.3576	0.0011				
Days Since Last Opened^2	-0.2961	0.0045				
Days Since Last Clicked			-0.0765	0.0064		
Days Since Last Clicked^2			0.0322	0.002		
Days Since Last Purchased	0.1443	0.0026	0.1228	0.014	-0.2327	0.0095
Days Since Last Purchased^2	-0.0147	0.0005	0.006	0.005	0.0725	0.0048

Table S.6: Stage-wise model estimates of NCL

Coefficients	Open stage		Click stage		Purchase stage	
	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	-0.8212	0.0028	-1.7773	0.0013	-4.2676	0.0049
Ave. Retail Spend	0.0006	0.0002	0.0004	0.0002	0.0009	0.0004
Ave. Web Spend	-0.0003	0.0002	0.0007	0.0002	0.0092	0.0005
Pur. Freq.	0.0032	0.0008	0.0124	0.0006	0.0423	0.002
Age	0.0813	0.0012	0.0739	0.0051	0.2916	0.0228
Income: 15K - 25K	0.0026	0.0022	-0.0046	0.0016	-0.0367	0.0038
Income: 25K - 35K	0.024	0.002	-0.0225	0.0016	-0.1065	0.002
Income: 35K - 50K	0.0272	0.0023	-0.0422	0.0008	-0.0226	0.0029
Income: 50K - 75K	0.0385	0.0035	-0.0454	0.0009	-0.0595	0.0015
Income: 75K - 100K	0.0518	0.002	-0.0617	0.001	-0.1378	0.002
Income: 100K - 120K	0.073	0.0016	-0.0671	0.0026	-0.1319	0.002
Income: 120K - 149K	0.0733	0.002	-0.0775	0.0013	0.0068	0.0042
Income: 150K plus	0.0978	0.0019	-0.0726	0.0012	0.0882	0.0013
Days Since Last Opened	1.6208	0.0049				
Days Since Last Opened^2	-0.6053	0.007				
Open: Knot 1	-1.4695	0.0052				
Open: Knot 2	2.9845	0.0074				
Open: Knot 3	-1.0986	0.0044				
Open: Knot 4	0.6465	0.0048				
Open: Knot 5	-1.0992	0.0046				
Open: Knot 6	0.8947	0.0039				
Days Since Last Clicked			-0.7544	0.0069		
Days Since Last Clicked^2			1.3947	0.0047		
Click: Knot 1			-2.584	0.0044		
Click: Knot 2			1.3811	0.0094		
Click: Knot 3			0.4162	0.0043		
Click: Knot 4			-1.231	0.0063		
Click: Knot 5			0.4002	0.0067		
Click: Knot 6			0.2666	0.0039		
Days Since Last Purchased	-0.5232	0.0031	0.0656	0.0088	-1.6554	0.0162
Days Since Last Purchased^2	0.0706	0.0052	-0.244	0.0049	1.5163	0.0096
Purchase: Knot 1	0.4417	0.006	0.8402	0.0102	-1.5476	0.0216
Purchase: Knot 2	-0.6094	0.0056	-0.7288	0.0086	-0.1684	0.0136
Purchase: Knot 3	0.1934	0.0034	-0.0431	0.0054	0.2146	0.0141
Purchase: Knot 4	-0.1801	0.0068	0.2862	0.0101	1.4211	0.0213
Purchase: Knot 5	0.0346	0.0064	-0.4838	0.01	-4.764	0.0151
Purchase: Knot 6	0.0358	0.0072	0.3721	0.0136	3.9507	0.0045

Table S.7: Stage-wise model estimates of CRON

Coefficients	Open stage		Click stage		Purchase stage	
	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	5.2852	0.0012	5.9916	0.0006	2.4787	0.0027
Ave. Retail Spend	0.0007	0.0001	-0.0003	1.00E-04	-0.0006	0.0002
Ave. Web Spend	-0.0003	1.00E-04	-0.0007	1.00E-04	0.0034	0.0002
Pur. Freq.	0.0035	0.0005	0.0060	0.0003	0.0233	0.0005
Age	0.0797	0.0009	-0.0360	0.0008	0.0877	0.0016
Income: 15K - 25K	-0.0078	0.0022	-0.0041	0.0017	-0.0877	0.0036
Income: 25K - 35K	0.0222	0.0008	-0.0621	0.0011	-0.0433	0.0014
Income: 35K - 50K	0.0346	0.0014	-0.0857	0.0009	0.0005	0.0019
Income: 50K - 75K	0.0310	0.0014	-0.1053	0.0006	0.0002	0.0020
Income: 75K - 100K	0.0581	0.0007	-0.1453	0.0008	0.0390	0.0022
Income: 100K - 120K	0.0812	0.0006	-0.1805	0.001	0.0464	0.0031
Income: 120K - 149K	0.0806	0.0007	-0.1960	0.0009	0.0936	0.0048
Income: 150K plus	0.1052	0.0006	-0.2229	0.0008	0.0845	0.0049
Days Since Last Opened	-14.6200	0.0016				
Days Since Last Opened^2	10.6154	0.0028				
Open: Knot 1	-12.3773	0.0016				
Open: Knot 2	2.8874	0.0020				
Open: Knot 3	-1.7181	0.0015				
Open: Knot 4	1.3061	0.0020				
Open: Knot 5	-1.5721	0.0017				
Open: Knot 6	1.1711	0.0020				
Days Since Last Clicked			-17.1054	0.0015		
Days Since Last Clicked^2			11.8874	0.0011		
Click: Knot 1			-11.6616	0.0014		
Click: Knot 2			-0.1183	0.0009		
Click: Knot 3			0.5473	0.0012		
Click: Knot 4			-1.4043	0.0012		
Click: Knot 5			0.7620	0.0017		
Click: Knot 6			-0.0432	0.0011		
Days Since Last Purchased	-0.4596	0.0025	-0.3902	0.0013	-13.9191	0.0020
Days Since Last Purchased^2	-0.0439	0.0008	0.0298	0.0015	10.0910	0.0015
Purchase: Knot 1	0.6753	0.0011	0.4361	0.0012	-10.1418	0.0024
Purchase: Knot 2	-0.7802	0.0007	-0.6298	0.0033	-0.0325	0.0020
Purchase: Knot 3	0.3035	0.0022	0.1602	0.0013	0.1334	0.0018
Purchase: Knot 4	-0.2311	0.0010	0.1646	0.0009	0.4342	0.0019
Purchase: Knot 5	0.0058	0.0011	-0.4595	0.0015	-1.8808	0.0044
Purchase: Knot 6	0.0600	0.0014	0.3763	0.0012	1.8666	0.0011
Promo 2	0.2537	0.0025	0.8472	0.0007	0.7154	0.0012
Promo 3	-6.0246	0.0010	-5.8443	0.0012	-3.8129	0.0012
Promo 4	0.1981	0.0009	0.1563	0.0011	-0.3890	0.0022
Promo 5	0.1501	0.0011	0.4743	0.0008	-0.1915	0.0016
Promo 6	-0.2022	0.0010	0.6781	0.0006	0.0814	0.0012
Promo 7	-0.3040	0.0015	-0.1519	0.0008	-0.2666	0.0012
Promo 8	-0.2167	0.0009	-0.1637	0.0006	0.3096	0.0011
Promo 9	-0.1851	0.0013	0.4961	0.0014	-0.1414	0.0017
Promo 10	0.3180	0.0019	0.6597	0.0005	0.1650	0.0025
Promo 11	-0.0628	0.0011	0.6766	0.0022	-0.3893	0.0023
Promo 12	-0.3803	0.0010	0.4773	0.0008	0.1363	0.0013
Promo 13	0.5475	0.0010	1.3002	0.0011	-0.4129	0.0018
Promo 14	-0.2244	0.0011	0.2209	0.0015	0.5437	0.0016
Promo 15	-6.0804	0.0014	-6.0259	0.0008	-3.8896	0.0008
Promo 16	0.3712	0.0012	0.6348	0.0006	-0.1371	0.0020
Promo 17	0.4937	0.0016	0.0979	0.0011	0.0820	0.0025
Promo 18	-0.2202	0.0009	-0.0051	0.0022	-0.0492	0.0034
Promo 19	-6.2357	0.0010	-6.4296	0.0011	-5.5452	0.0018
Promo 20	0.3035	0.0008	0.4395	0.0007	0.3001	0.0014
Promo 21	0.4908	0.0014	0.2896	0.0012	0.4204	0.0014
Promo 22	0.4742	0.0013	0.1394	0.0009	0.2866	0.0020
Promo 23	-6.0384	0.0009	-6.0974	0.0015	-3.9750	0.0033
Promo 24	-0.2334	0.0024	0.1113	0.0007	-2.5912	0.0013
Promo 25	0.1151	0.0017	0.7958	0.0008	-0.7408	0.0014

S.4 Details on the SCRON model

S.4.1 Hyper-parameter Calibration

For the SCRON model we had tried several hyper-parameter choices before selecting the chosen hyper-parameters. The form of spike and slab prior is $\pi_\eta = (1-\eta)N(0,\tau_1) + \eta N(0,\tau_2)$, where $\tau_2 = \tau_1/\kappa$. For hyper-parameter calibration, we restrict ourselves to an arbitrary ($L = 1$) subsample of the data. As joint calibration of the hyper-parameters would be very tedious, we use greedy methods and heuristics for calibration.

Initially, we set $\eta = 0$ and using a non-informative prior on τ_1, i.e., $\pi(\tau_1) \propto 1$, we conduct Bayesian estimation based on the BMH algorithm. Once the MCMC iterates stabilizes, we calculate the posterior mean and variance of τ_1. We next considered an Inverse Gamma distribution on τ_1 whose mean and variance were selected in the following two ways (a) such that they well approximate the posterior mean and variance (b) based on the MLE of the Inverse Gamma distribution. We run the marginal models with both the choices and noticed that there is not much difference in training ROC curves. We chose the MLE estimates as parameters of the Inverse Gamma prior for τ_1.

Thereafter, we set the sparsity parameter η equal to non-response rate for the corresponding promotion:segment combination and consider selection of τ_2. Our choice of $\kappa = 1000$ is influenced by the procedure prescribed in Malsiner-Walli and Wagner [2011] for tuning spike-and-slab based estimation methodology with absolutely continuous spikes. We also tried different values of κ, viz., 500, 750, 1000, 1250, 1500, 2000 and compared the training ROC curves. We found $\kappa = 750$ to 1250 produced similar curves which are better than the fits from other choices. So, we set $\kappa = 1000$. With the prior on τ_1 and the value of κ set, we next consider a Beta prior on η whose mean equals the corresponding non-response rate. Care was taken to have considerable variance in these Beta priors by keeping the α-shape parameter fixed and set to a small value while the β-shape varies with the non-response rate.

S.4.2 Data summary of the High, Mid and Low engaged customer groups

Tables S.8–S.10 document the average attributes of the three customer segments considered in Sections 5.4 and 5.5 of the main paper.

Table S.8: Summary statistics corresponding to the high-engaged segments of customers studied via SCRON model in Section 5.5 of the main paper

Promotions			Responses			Days: Last Open		Days: Last Click		Days: Last Purchased		Pur. Freq.		Avg. Retail Spend		Avg. Web Spend	
ID No	Type No	Recipients	Opened	Clicked	Ordered	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1	N	46245	0.3749	0.1114	0.013	7.78	15.97	21.53	84.26	50.24	86.39	9.35	9.99	14.98	30.93	52.08	32.00
2	P	46623	0.458	0.2367	0.0507	7.38	15.80	20.83	82.72	48.61	83.16	9.35	9.99	14.97	30.90	52.07	32.01
3	P	47999	0.3979	0.1778	0.0195	0.33	4.14	1.74	25.80	18.56	34.03	9.59	9.98	15.00	30.86	52.45	31.52
4	N	46913	0.4654	0.1531	0.0124	6.55	14.35	18.00	77.28	40.67	54.39	9.34	9.98	14.97	30.90	52.03	32.02
5	P	36867	0.4653	0.189	0.0167	5.25	10.15	12.90	52.18	36.11	47.78	9.50	9.97	15.13	30.94	52.48	32.01
6	C	46915	0.4029	0.1834	0.0217	4.62	9.38	10.69	42.89	33.98	44.83	9.52	9.99	15.10	30.97	52.40	31.84
7	N	46748	0.3466	0.0889	0.0083	5.56	11.30	13.66	56.79	37.40	50.13	9.51	9.98	15.09	30.97	52.42	31.85
8	N	23716	0.4035	0.1084	0.0156	5.69	12.47	15.41	64.90	41.12	53.77	9.37	10.07	14.82	30.89	52.37	32.25
9	C	46898	0.3768	0.1548	0.0156	6.29	13.23	17.13	70.14	40.21	53.74	9.34	9.98	14.97	30.90	52.03	32.02
10	P	9387	0.4409	0.2021	0.0255	7.01	13.80	16.66	57.76	42.08	52.98	9.39	9.88	15.10	31.56	52.38	31.42
11	C	47756	0.4114	0.1823	0.013	4.30	5.84	7.83	26.77	30.31	34.69	9.60	9.99	14.97	30.83	52.48	31.49
12	C	46907	0.3732	0.1519	0.0189	4.36	8.80	9.77	41.30	34.97	44.80	9.52	9.99	15.09	30.97	52.41	31.84
13	P	47920	0.549	0.3379	0.0245	4.62	7.43	9.10	36.17	32.89	41.23	9.59	9.97	14.99	30.86	52.47	31.51
14	P	47590	0.3614	0.1232	0.0212	4.66	7.07	9.20	34.09	32.40	40.33	9.58	9.97	15.01	30.88	52.47	31.51
15	P	24137	0.3357	0.1078	0.0058	0.60	2.97	2.06	9.06	22.36	27.14	9.95	10.09	15.00	30.74	52.73	30.75
16	p	46623	0.426	0.1813	0.0167	5.34	5.84	8.84	26.43	31.55	33.50	9.72	9.95	15.07	30.87	52.60	31.01
17	N	22796	0.5492	0.1656	0.0161	4.58	4.73	8.24	21.55	31.95	31.33	7.46	7.62	13.35	29.12	51.03	31.08
18	C	47368	0.3514	0.0987	0.0096	4.64	5.46	7.96	24.82	29.60	30.75	9.78	10.01	15.04	30.84	52.56	30.98
19	C	46916	0.3246	0.0829	0.0016	0.27	1.55	1.28	3.23	19.51	23.83	10.05	10.10	15.04	30.60	52.25	30.51
20	C	21079	0.3905	0.1435	0.0185	5.46	4.28	9.54	8.65	31.62	27.59	8.44	9.18	13.33	29.34	52.04	31.43
21	N	47206	0.368	0.1212	0.0171	6.31	4.15	10.29	7.50	31.27	26.40	9.92	10.05	15.09	30.78	52.48	30.85
22	N	47522	0.5406	0.1704	0.0217	4.52	4.68	8.04	16.48	30.93	29.07	9.80	10.02	15.06	30.84	52.53	31.01
23	C	23504	0.3522	0.1065	0.0081	0.55	2.44	2.58	5.00	22.36	23.77	10.18	10.10	15.09	30.39	52.18	30.41
24	P	31467	0.3765	0.1079	0.0008	4.20	4.37	8.26	8.35	30.97	26.37	8.44	8.24	16.10	31.87	54.08	31.97
25	C	47269	0.4533	0.2107	0.0102	4.20	4.29	8.07	7.74	30.16	27.01	9.91	10.05	15.09	30.78	52.47	30.84

Table S.9: Summary statistics corresponding to the mid-engaged segments of customers studied via SCRON model in Section 5.5 of the main paper

Promotions			Responses			Days: Last Open		Days: Last Click		Days: Last Purchased		Pur. Freq.		Avg. Retail Spend		Avg. Web Spend	
ID No	Type No	Recipients	Opened	Clicked	Ordered	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1	N	21042	0.3368	0.0895	0.0011	11.85	21.56	33.86	100.55	115.56	35.54	5.73	5.26	13.06	30.09	53.17	36.38
2	P	21083	0.398	0.1705	0.0034	11.50	21.59	33.32	99.98	116.42	35.72	5.74	5.26	13.06	30.08	53.15	36.37
3	P	21455	0.3617	0.1395	0.0032	1.38	10.28	6.05	46.29	105.93	49.26	5.66	5.18	12.87	29.88	53.17	36.37
4	N	21157	0.4156	0.1324	0.0026	10.59	20.92	30.09	94.31	117.62	36.78	5.74	5.26	13.06	30.06	53.14	36.35
5	P	16593	0.419	0.1533	0.0044	8.12	17.43	22.49	74.41	120.56	40.51	5.71	5.24	13.03	29.14	53.10	36.35
6	C	21091	0.3915	0.1689	0.0066	7.20	16.11	19.11	64.46	120.57	43.36	5.71	5.22	13.00	29.96	53.12	36.34
7	N	21028	0.3316	0.0848	0.0014	8.66	18.00	23.77	78.01	119.71	39.35	5.70	5.21	12.99	29.97	53.14	36.35
8	N	10757	0.3561	0.0886	0.0023	9.25	19.81	26.92	92.94	119.13	37.61	5.71	5.33	12.96	29.27	53.32	36.66
9	C	21153	0.3546	0.1353	0.0021	10.19	20.29	29.43	93.02	118.04	37.61	5.74	5.26	13.06	30.06	53.14	36.35
10	P	4271	0.3706	0.1496	0.0037	10.85	21.22	28.29	82.88	120.41	37.27	5.70	5.24	12.58	28.97	52.75	34.24
11	C	21370	0.4027	0.1727	0.0056	6.04	12.21	13.08	47.63	119.98	51.23	5.66	5.18	12.87	29.90	53.16	36.31
12	C	21086	0.3649	0.1414	0.0056	6.81	15.60	17.82	62.53	121.57	43.36	5.71	5.22	12.99	29.96	53.12	36.34
13	P	21435	0.4972	0.2782	0.0074	6.73	14.20	15.96	56.61	121.37	45.85	5.66	5.18	12.88	29.89	53.17	36.38
14	P	21282	0.3495	0.1078	0.0084	6.67	13.91	15.84	56.08	121.29	46.90	5.66	5.18	12.87	29.90	53.19	36.37
15	P	10624	0.3273	0.1057	0.0024	1.02	5.57	3.40	13.98	111.68	57.27	5.74	5.16	12.94	30.43	53.80	37.91
16	p	20876	0.4169	0.173	0.0085	6.97	11.79	13.71	46.46	121.79	51.26	5.67	5.15	12.90	30.11	53.32	36.29
17	N	11739	0.4945	0.1489	0.0087	5.42	8.65	10.37	21.37	122.26	57.24	4.79	4.23	11.45	27.72	51.23	35.01
18	C	21171	0.3509	0.0985	0.0052	6.01	11.09	11.78	37.22	119.74	54.11	5.68	5.17	12.89	30.06	53.30	36.22
19	C	20909	0.3366	0.0928	0.001	0.35	1.76	1.36	3.37	99.71	66.84	5.78	5.16	12.92	30.12	53.46	35.92
20	C	10354	0.4006	0.1451	0.0097	6.17	6.77	10.56	16.60	120.08	61.92	4.89	4.59	10.88	27.27	52.18	36.61
21	N	21053	0.3722	0.1241	0.011	6.94	6.46	11.31	13.94	117.19	63.71	5.73	5.16	12.88	30.05	53.42	36.27
22	N	21224	0.4968	0.1522	0.0123	5.33	8.46	9.93	20.18	119.58	58.54	5.69	5.17	12.92	30.08	53.28	36.31
23	C	10466	0.3728	0.1214	0.0066	0.69	2.75	2.68	5.09	101.54	69.79	5.83	5.09	12.71	29.57	53.86	35.54
24	P	14780	0.3842	0.1225	0.0007	4.95	7.55	9.65	16.78	119.46	60.39	5.10	4.59	13.88	31.24	54.38	36.87
25	C	21103	0.4611	0.2131	0.0069	4.92	7.01	9.35	15.26	117.71	61.31	5.72	5.17	12.87	30.02	53.39	36.20

Table S.10: Summary statistics corresponding to the low-engaged segments of customers studied via SCRON model in Section 5.5 of the main paper

Promotions			Responses			Days: Last Open		Days: Last Click		Days: Last Purchased		Pur. Freq.		Avg. Retail Spend		Avg. Web Spend	
ID No	Type No	Recipients	Opened	Clicked	Ordered	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1	N	8374	0.3262	0.091	0.0011	15.19	33.53	43.00	100.92	275.89	76.53	3.43	3.04	10.61	27.44	49.35	36.65
2	P	8397	0.3942	0.173	0.0051	14.87	33.47	42.31	99.41	276.42	76.17	3.43	3.04	10.62	27.44	49.32	36.60
3	P	8532	0.3511	0.1326	0.0026	2.05	15.77	8.56	50.21	255.91	84.88	3.39	2.98	10.52	27.26	49.32	36.45
4	N	8422	0.4064	0.1306	0.0007	13.65	32.59	38.09	93.71	273.72	53.63	3.43	3.04	10.60	27.41	49.32	36.63
5	P	6588	0.3956	0.1439	0.0038	10.02	26.86	27.88	76.38	276.49	69.35	3.40	2.98	10.45	27.36	49.26	36.25
6	C	8392	0.3787	0.1612	0.0048	8.87	25.42	23.88	73.50	274.08	75.35	3.41	3.00	10.58	27.41	49.38	36.58
7	N	8360	0.3283	0.0911	0.002	10.72	27.98	29.98	84.13	275.71	65.15	3.41	3.00	10.57	27.35	49.37	36.56
8	N	4309	0.3483	0.0907	0.0023	11.79	30.89	33.44	91.42	275.78	55.17	3.41	3.03	10.40	27.13	49.26	36.84
9	C	8419	0.3403	0.1378	0.0017	13.12	31.73	36.96	90.90	273.89	55.60	3.43	3.04	10.60	27.41	49.32	36.64
10	P	1774	0.3653	0.1471	0.0034	12.45	30.03	33.74	65.88	277.41	54.24	3.36	2.89	11.65	30.50	49.60	35.56
11	C	8014	0.3868	0.1693	0.0044	6.93	17.79	16.19	53.10	264.14	85.10	3.43	3.01	10.59	27.46	49.31	36.32
12	C	8392	0.3509	0.135	0.0039	8.46	24.42	22.85	72.93	275.04	75.41	3.41	3.01	10.59	27.41	49.37	36.58
13	P	8519	0.4692	0.2597	0.0055	8.28	22.86	20.25	67.22	274.57	74.13	3.39	2.98	10.51	27.24	49.34	36.46
14	P	8451	0.3374	0.1045	0.0052	8.12	22.27	19.99	66.10	273.59	77.25	3.39	2.98	10.46	27.20	49.31	36.44
15	P	4047	0.3279	0.1016	0.0025	1.86	13.66	5.74	45.52	250.58	100.08	3.47	3.07	10.74	27.83	49.89	36.99
16	p	7822	0.4118	0.1718	0.0063	7.91	17.67	16.84	52.45	264.89	86.19	3.44	3.00	10.52	27.47	49.76	36.70
17	N	5416	0.4629	0.1405	0.0063	6.20	13.11	12.24	36.57	266.97	100.65	3.08	2.71	9.78	26.58	48.68	35.73
18	C	7864	0.3472	0.1055	0.0028	7.04	17.36	15.01	50.76	259.89	92.66	3.45	3.01	10.56	27.49	49.71	36.68
19	C	8320	0.3376	0.1076	0.0006	0.40	1.89	1.43	3.46	238.48	122.17	3.45	2.96	10.53	27.30	49.87	36.57
20	C	4643	0.3853	0.1495	0.0108	6.77	10.12	12.16	22.65	264.75	108.37	3.04	2.61	9.36	25.30	48.41	36.10
21	N	8374	0.3719	0.1266	0.0085	7.57	9.90	12.46	18.69	259.87	113.17	3.42	2.97	10.48	27.33	49.68	36.68
22	N	8431	0.4728	0.14	0.0075	5.97	12.52	11.69	38.61	265.01	102.99	3.40	2.97	10.51	27.28	49.61	36.72
23	C	3785	0.3482	0.1184	0.0045	0.87	3.08	2.91	5.27	231.76	122.65	3.51	2.95	10.60	26.39	50.18	36.33
24	P	5995	0.3708	0.1216	0.0005	5.49	10.99	10.80	21.20	263.30	106.67	3.18	2.72	11.31	28.81	50.70	37.67
25	C	8389	0.4452	0.2011	0.005	5.50	10.56	10.54	20.52	262.06	108.21	3.41	2.97	10.46	27.31	49.64	36.64

We do not see much difference in the coverage of the promotions or the mean of covariates among groups. In these scenario, we will not have any issue in estimation of the SCRON model. However, for customer segments with smaller sizes (for example for some rare segments) all promotion effects might not be well-estimated and so, such analysis will need more careful introspection than naive application of the SCRON model as done here.

S.4.3 Fitted model coefficients and standard errors

In Figure S.2, and in Tables S.11, S.12, and S.13 present the coefficient of SCRON. Since we have 25*3=75 levels of (promotion * engagement) interactions, which creates 74 dummy variables, with Promo ID 1 of low-engagement segment being the baseline.So, Promo ID 1 is in Tables S.12 and S.13 but not in Table S.11. In Figure S.2, the respective intercepts were added to the coefficients.

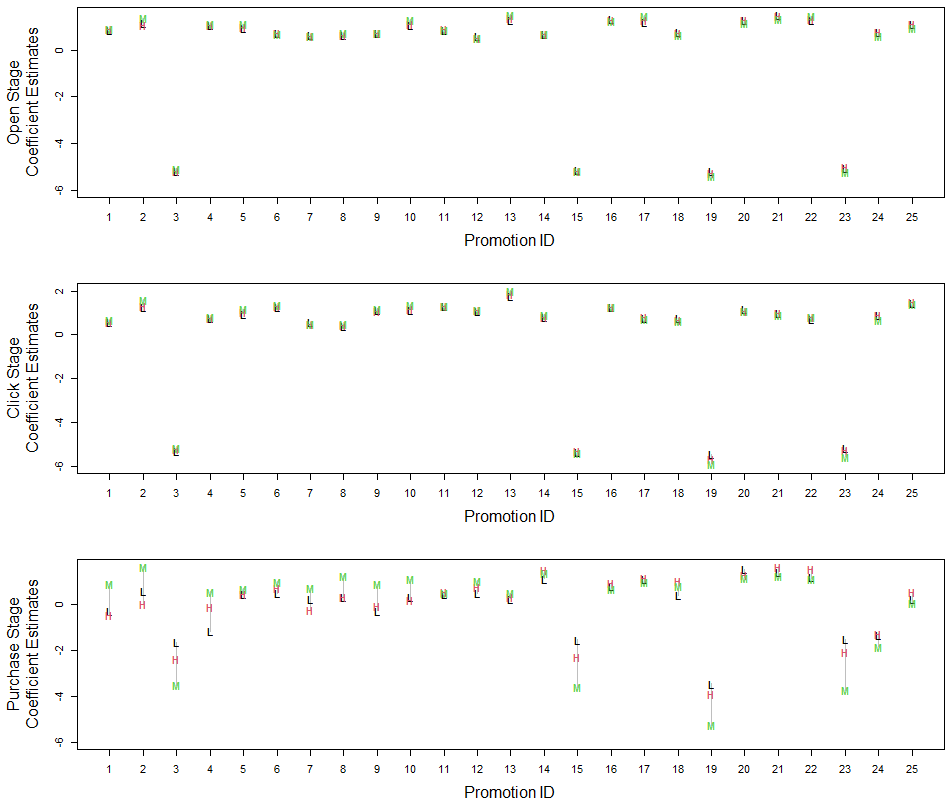


Figure S.2: Plot of estimates of the promotion effects from SCRON model. In the subplots from top to bottom, we have estimates from the open, click, and purchase stages respectively. H, M, and L, respectively, denote the high, mid, and low engaged customers.

Table S.11: Stage-wise model estimates of SCRON for low engagement customers

Coefficients	Open stage		Click stage		Purchase stage	
	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	5.1055	0.0023	5.7398	0.0010	0.0189	0.0024
Ave. Retail Spend	0.0006	0.0002	-0.0003	0.0002	-0.0007	0.0003
Ave. Web Spend	-0.0002	0.0002	-0.0007	0.0002	0.0040	0.0003
Pur. Freq.	0.0032	0.0006	0.0056	0.0008	0.0236	0.0009
Age	0.0759	0.0012	-0.0354	0.0009	0.1212	0.0027
Income: 15K - 25K	-0.0026	0.0018	-0.0102	0.0015	-0.0563	0.0012
Income: 25K - 35K	0.0204	0.0017	-0.0647	0.0019	-0.0461	0.0029
Income: 35K - 50K	0.0294	0.0014	-0.0862	0.0007	0.0133	0.0015
Income: 50K - 75K	0.0415	0.0020	-0.0980	0.0023	0.0229	0.0029
Income: 75K - 100K	0.0576	0.0024	-0.1431	0.0021	0.0710	0.0009
Income: 100K - 120K	0.0744	0.0019	-0.1760	0.0010	0.0266	0.0012
Income: 120K - 149K	0.0823	0.0018	-0.1888	0.0015	0.1005	0.0015
Income: 150K plus	0.0999	0.0009	-0.2264	0.0006	0.0806	0.0015
Days Since Last Opened	-14.6378	0.0092				
Days Since Last Opened^2	10.6483	0.0114				
Open: Knot 1	-12.4159	0.0055				
Open: Knot 2	2.8864	0.0068				
Open: Knot 3	-1.7323	0.0100				
Open: Knot 4	1.3113	0.0092				
Open: Knot 5	-1.5789	0.0065				
Open: Knot 6	1.1663	0.0058				
Days Since Last Clicked			-17.1010	0.0097		
Days Since Last Clicked^2			11.8783	0.0059		
Click: Knot 1			-11.6513	0.0079		
Click: Knot 2			-0.1214	0.0029		
Click: Knot 3			0.5692	0.0042		
Click: Knot 4			-1.3804	0.0021		
Click: Knot 5			0.6875	0.0056		
Click: Knot 6			-0.0211	0.0060		
Days Since Last Purchased	-0.4478	0.0053	-0.4946	0.0043	-15.3012	0.0258
Days Since Last Purchased^2	-0.0714	0.0038	0.0923	0.0029	11.1235	0.0158
Purchase: Knot 1	0.7259	0.0063	0.3957	0.0031	-11.4026	0.0243
Purchase: Knot 2	-0.8141	0.0049	-0.6642	0.0049	-0.1895	0.0193
Purchase: Knot 3	0.3235	0.0054	0.2242	0.0039	-0.0839	0.0043
Purchase: Knot 4	-0.2684	0.0151	0.0393	0.0059	4.6886	0.0133
Purchase: Knot 5	0.2064	0.0033	-0.1370	0.0059	-7.8951	0.0356
Purchase: Knot 6	-0.1359	0.0046	0.0900	0.0037	3.7391	0.0189
Promo 2	0.2631	0.0028	0.6849	0.0007	0.8665	0.0011
Promo 3	-6.0722	0.0027	-5.8700	0.0012	-1.3692	0.0011
Promo 4	0.2027	0.0014	0.1776	0.0010	-0.8826	0.0021
Promo 5	0.0641	0.0015	0.3807	0.0012	0.7319	0.0012
Promo 6	-0.1297	0.0021	0.6755	0.0010	0.7572	0.0009
Promo 7	-0.2311	0.0011	0.0000	0.0000	0.5025	0.0011
Promo 8	-0.2476	0.0016	-0.1639	0.0011	0.6080	0.0025
Promo 9	-0.1492	0.0009	0.5648	0.0016	0.0000	0.0000
Promo 10	0.1990	0.0027	0.5603	0.0010	0.5918	0.0026
Promo 11	0.0000	0.0000	0.7477	0.0013	0.7340	0.0034
Promo 12	-0.2903	0.0014	0.5212	0.0015	0.7696	0.0010
Promo 13	0.4271	0.0014	1.1998	0.0009	0.5260	0.0013
Promo 14	-0.1899	0.0014	0.2088	0.0010	1.3859	0.0008
Promo 15	-5.9961	0.0014	-5.9491	0.0015	-1.2487	0.0014
Promo 16	0.4458	0.0012	0.6753	0.0010	1.0777	0.0015
Promo 17	0.3310	0.0018	0.2040	0.0012	1.3672	0.0027
Promo 18	-0.0938	0.0015	0.1729	0.0006	0.6785	0.0045
Promo 19	-6.0752	0.0009	-6.0400	0.0005	-3.1790	0.0010
Promo 20	0.4016	0.0028	0.5986	0.0013	1.8092	0.0012
Promo 21	0.6212	0.0018	0.4186	0.0017	1.7045	0.0014
Promo 22	0.4019	0.0038	0.1374	0.0010	1.4751	0.0032
Promo 23	-5.9312	0.0010	-5.7605	0.0011	-1.2423	0.0028
Promo 24	-0.1178	0.0014	0.3373	0.0007	-1.0473	0.0014
Promo 25	0.2477	0.0017	0.8613	0.0009	0.5016	0.0019

Table S.12: Stage-wise model estimates of SCRON for mid engagement customers

Coefficients	Open stage		Click stage		Purchase stage	
	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	5.1055	0.0023	5.7398	0.0010	0.0189	0.0024
Ave. Retail Spend	0.0006	0.0002	-0.0003	0.0002	-0.0007	0.0003
Ave. Web Spend	-0.0002	0.0002	-0.0007	0.0002	0.0040	0.0003
Pur. Freq.	0.0032	0.0006	0.0056	0.0008	0.0236	0.0009
Age	0.0759	0.0012	-0.0354	0.0009	0.1212	0.0027
Income: 15K - 25K	-0.0026	0.0018	-0.0102	0.0015	-0.0563	0.0012
Income: 25K - 35K	0.0204	0.0017	-0.0647	0.0019	-0.0461	0.0029
Income: 35K - 50K	0.0294	0.0014	-0.0862	0.0007	0.0133	0.0015
Income: 50K - 75K	0.0415	0.0020	-0.0980	0.0023	0.0229	0.0029
Income: 75K - 100K	0.0576	0.0024	-0.1431	0.0021	0.0710	0.0009
Income: 100K - 120K	0.0744	0.0019	-0.1760	0.0010	0.0266	0.0012
Income: 120K - 149K	0.0823	0.0018	-0.1888	0.0015	0.1005	0.0015
Income: 150K plus	0.0999	0.0009	-0.2264	0.0006	0.0806	0.0015
Days Since Last Opened	-14.6378	0.0092				
Days Since Last Opened^2	10.6483	0.0114				
Open: Knot 1	-12.4159	0.0055				
Open: Knot 2	2.8864	0.0068				
Open: Knot 3	-1.7323	0.0100				
Open: Knot 4	1.3113	0.0092				
Open: Knot 5	-1.5789	0.0065				
Open: Knot 6	1.1663	0.0058				
Days Since Last Clicked			-17.1010	0.0097		
Days Since Last Clicked^2			11.8783	0.0059		
Click: Knot 1			-11.6513	0.0079		
Click: Knot 2			-0.1214	0.0029		
Click: Knot 3			0.5692	0.0042		
Click: Knot 4			-1.3804	0.0021		
Click: Knot 5			0.6875	0.0056		
Click: Knot 6			-0.0211	0.0060		
Days Since Last Purchased	-0.4478	0.0053	-0.4946	0.0043	-15.3012	0.0258
Days Since Last Purchased^2	-0.0714	0.0038	0.0923	0.0029	11.1235	0.0158
Purchase: Knot 1	0.7259	0.0063	0.3957	0.0031	-11.4026	0.0243
Purchase: Knot 2	-0.8141	0.0049	-0.6642	0.0049	-0.1895	0.0193
Purchase: Knot 3	0.3235	0.0054	0.2242	0.0039	-0.0839	0.0043
Purchase: Knot 4	-0.2684	0.0151	0.0393	0.0059	4.6886	0.0133
Purchase: Knot 5	0.2064	0.0033	-0.137	0.0059	-7.8951	0.0356
Purchase: Knot 6	-0.1359	0.0046	0.0900	0.0037	3.7391	0.0189
Promo 1	0.0600	0.0022	0.0660	0.0011	0.2869	0.0015
Promo 2	0.2631	0.0028	0.8079	0.0032	0.7316	0.0032
Promo 3	-6.0018	0.0042	-5.7259	0.0022	-1.6511	0.0030
Promo 4	0.2370	0.0036	0.2756	0.0023	0.6119	0.0038
Promo 5	0.1710	0.0028	0.5393	0.0035	1.1760	0.0018
Promo 6	-0.0739	0.0036	0.8038	0.0029	1.4671	0.0036
Promo 7	-0.2311	0.0011	0.0000	0.0000	0.5025	0.0011
Promo 8	-0.1854	0.0039	-0.0561	0.0028	1.0623	0.0054
Promo 9	-0.0918	0.0031	0.5648	0.0016	0.6738	0.0014
Promo 10	0.2966	0.0039	0.6717	0.0017	0.9202	0.0035
Promo 11	0.0669	0.0009	0.8296	0.0029	1.2507	0.0059
Promo 12	-0.2903	0.0014	0.6151	0.0030	1.4947	0.0022
Promo 13	0.5607	0.0026	1.3398	0.0017	1.0655	0.0031
Promo 14	-0.1246	0.0043	0.3207	0.0023	2.2080	0.0023
Promo 15	-5.9961	0.0014	-5.7910	0.0020	-1.5469	0.0033
Promo 16	0.4458	0.0012	0.7847	0.0026	1.6643	0.0037
Promo 17	0.4797	0.0039	0.2891	0.0024	1.8743	0.0039
Promo 18	-0.0938	0.0015	0.1729	0.0006	1.7653	0.0059
Promo 19	-6.0752	0.0009	-6.1621	0.0018	-3.1790	0.0010
Promo 20	0.4662	0.0047	0.5986	0.0013	1.9910	0.0024
Promo 21	0.6212	0.0018	0.4997	0.0038	2.3634	0.0031
Promo 22	0.4947	0.0059	0.3144	0.0027	2.2773	0.0049
Promo 23	-5.8418	0.0033	-5.7605	0.0011	-1.3279	0.0044
Promo 24	-0.0652	0.0036	0.3816	0.0020	-0.5405	0.0023
Promo 25	0.3012	0.0046	0.9940	0.0024	1.2706	0.0031

Table S.13: Stage-wise model estimates of SCRON for high engagement customers

Coefficients	Open stage		Click stage		Purchase stage	
	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	5.1055	0.0023	5.7398	0.0010	0.0189	0.0024
Ave. Retail Spend	0.0006	0.0002	-0.0003	0.0002	-0.0007	0.0003
Ave. Web Spend	-0.0002	0.0002	-0.0007	0.0002	0.0040	0.0003
Pur. Freq.	0.0032	0.0006	0.0056	0.0008	0.0236	0.0009
Age	0.0759	0.0012	-0.0354	0.0009	0.1212	0.0027
Income: 15K - 25K	-0.0026	0.0018	-0.0102	0.0015	-0.0563	0.0012
Income: 25K - 35K	0.0204	0.0017	-0.0647	0.0019	-0.0461	0.0029
Income: 35K - 50K	0.0294	0.0014	-0.0862	0.0007	0.0133	0.0015
Income: 50K - 75K	0.0415	0.0020	-0.0980	0.0023	0.0229	0.0029
Income: 75K - 100K	0.0576	0.0024	-0.1431	0.0021	0.0710	0.0009
Income: 100K - 120K	0.0744	0.0019	-0.1760	0.0010	0.0266	0.0012
Income: 120K - 149K	0.0823	0.0018	-0.1888	0.0015	0.1005	0.0015
Income: 150K plus	0.0999	0.0009	-0.2264	0.0006	0.0806	0.0015
Days Since Last Opened	-14.6378	0.0092	0	0	0	0
Days Since Last Opened^2	10.6483	0.0114	0	0	0	0
Open: Knot 1	-12.4159	0.0055	0	0	0	0
Open: Knot 2	2.8864	0.0068	0	0	0	0
Open: Knot 3	-1.7323	0.0100	0	0	0	0
Open: Knot 4	1.3113	0.0092	0	0	0	0
Open: Knot 5	-1.5789	0.0065	0	0	0	0
Open: Knot 6	1.1663	0.0058	0	0	0	0
Days Since Last Clicked	0	0	-17.1010	0.0097	0	0
Days Since Last Clicked^2	0	0	11.8783	0.0059	0	0
Click: Knot 1	0	0	-11.6513	0.0079	0	0
Click: Knot 2	0	0	-0.1214	0.0029	0	0
Click: Knot 3	0	0	0.5692	0.0042	0	0
Click: Knot 4	0	0	-1.3804	0.0021	0	0
Click: Knot 5	0	0	0.6875	0.0056	0	0
Click: Knot 6	0	0	-0.0211	0.0060	0	0
Days Since Last Purchased	-0.4478	0.0053	-0.4946	0.0043	-15.3012	0.0258
Days Since Last Purchased^2	-0.0714	0.0038	0.0923	0.0029	11.1235	0.0158
Purchase: Knot 1	0.7259	0.0063	0.3957	0.0031	-11.4026	0.0243
Purchase: Knot 2	-0.8141	0.0049	-0.6642	0.0049	-0.1895	0.0193
Purchase: Knot 3	0.3235	0.0054	0.2242	0.0039	-0.0839	0.0043
Purchase: Knot 4	-0.2684	0.0151	0.0393	0.0059	4.6886	0.0133
Purchase: Knot 5	0.2064	0.0033	-0.1370	0.0059	-7.8951	0.0356
Purchase: Knot 6	-0.1359	0.0046	0.0900	0.0037	3.7391	0.0189
Promo 1	0.2054	0.0015	0.3237	0.0022	3.0569	0.0025
Promo 2	0.6518	0.0035	1.2511	0.0019	3.8000	0.0028
Promo 3	-5.7769	0.0061	-5.5277	0.0021	-1.3692	0.0011
Promo 4	0.4282	0.0026	0.4634	0.0019	2.6974	0.0033
Promo 5	0.3987	0.0036	0.8169	0.0028	2.8405	0.0024
Promo 6	-0.0197	0.0033	0.9983	0.0020	3.1506	0.0022
Promo 7	-0.1136	0.0023	0.1235	0.0018	2.8620	0.0041
Promo 8	0.0118	0.0036	0.1383	0.0030	3.4015	0.0049
Promo 9	0.0136	0.0023	0.8328	0.0031	3.0284	0.0010
Promo 10	0.5951	0.0045	1.0205	0.0021	3.2504	0.0046
Promo 11	0.1377	0.0012	0.9701	0.0042	2.6549	0.0047
Promo 12	-0.2012	0.0041	0.7815	0.0030	3.1772	0.0021
Promo 13	0.8168	0.0030	1.6711	0.0031	2.6507	0.0041
Promo 14	-0.0296	0.0024	0.5644	0.0027	3.5224	0.0038
Promo 15	-5.8775	0.0023	-5.7710	0.0040	-1.4460	0.0028
Promo 16	0.5469	0.0038	0.9328	0.0020	2.8437	0.0029
Promo 17	0.7729	0.0040	0.3807	0.0039	3.1104	0.0039
Promo 18	-0.0374	0.0030	0.2739	0.0034	2.9723	0.0056
Promo 19	-6.0752	0.0009	-6.2444	0.0016	-3.0993	0.0027
Promo 20	0.4483	0.0045	0.7262	0.0020	3.3156	0.0029
Promo 21	0.6212	0.0018	0.5576	0.0030	3.3863	0.0029
Promo 22	0.7453	0.0072	0.4492	0.0025	3.2637	0.0054
Promo 23	-5.9312	0.0010	-5.9169	0.0018	-1.5900	0.0044
Promo 24	-0.1178	0.0014	0.3373	0.0007	0.3053	0.0031
Promo 25	0.2477	0.0017	1.0814	0.0033	2.2192	0.0037

S.4.4 Comparing SCRON with other joint models

Table S.14: For different levels of false positive rates, we report the correct detection rates of conversions in the test data from the ROC curves of four models in Figure 4.

Data Type	Stage	Model	False Positive Rate							
			0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Training	Open	NCG	0.47	0.58	0.68	0.77	0.83	0.89	0.93	0.97
		NCL	0.44	0.57	0.68	0.77	0.83	0.89	0.93	0.97
		CRON	0.47	0.6	0.7	0.78	0.84	0.9	0.94	0.98
		SCRON	0.47	0.6	0.7	0.78	0.85	0.9	0.94	0.98
	Click	NCG	0.21	0.32	0.42	0.52	0.62	0.72	0.82	0.91
		NCL	0.27	0.37	0.47	0.56	0.65	0.74	0.82	0.91
		CRON	0.34	0.46	0.56	0.66	0.75	0.83	0.89	0.96
		SCRON	0.35	0.46	0.57	0.66	0.75	0.83	0.9	0.96
	Purchase	NCG	0.29	0.41	0.51	0.61	0.7	0.79	0.87	0.94
		NCL	0.31	0.43	0.54	0.64	0.73	0.81	0.88	0.95
		CRON	0.44	0.56	0.67	0.76	0.85	0.92	0.97	0.998
		SCRON	0.46	0.58	0.7	0.8	0.88	0.95	0.98	0.998
Test	Open	NCG	0.36	0.5	0.62	0.73	0.82	0.88	0.93	0.97
		NCL	0.42	0.55	0.65	0.74	0.82	0.88	0.93	0.97
		CRON	0.45	0.59	0.69	0.77	0.84	0.9	0.94	0.98
		SCRON	0.44	0.59	0.69	0.77	0.84	0.9	0.94	0.98
	Click	NCG	0.2	0.3	0.4	0.5	0.61	0.7	0.8	0.9
		NCL	0.23	0.35	0.45	0.55	0.65	0.72	0.81	0.9
		CRON	0.33	0.45	0.55	0.65	0.74	0.83	0.89	0.96
		SCRON	0.33	0.45	0.55	0.65	0.75	0.83	0.9	0.97
	Purchase	NCG	0.26	0.38	0.49	0.58	0.67	0.76	0.84	0.92
		NCL	0.29	0.4	0.51	0.61	0.7	0.78	0.86	0.93
		CRON	0.45	0.55	0.65	0.74	0.82	0.89	0.96	0.998
		SCRON	0.47	0.6	0.71	0.81	0.88	0.94	0.98	0.998

S.4.5 Comparing SCRON with Random Forest and XGBoost

We compare the predictive performance of our proposed joint model, SCRON, with the state-of-the-art machine learning algorithms such as Random Forest (RF) and XGBoost (XGB). We used the same training data and the same set of variables, as in our SCRON model, to train three unconditional RF models and likewise three unconditional XGB models. To optimize performance, the RF models were tuned over the parameters of

number of trees, max number of levels in each decision tree, min number of data points placed in a node before the node is split and max number of features considered for splitting a node training. The XGB models were tuned over the parameters like maximum number of boosting iterations, maximum depth of a tree and subsample ratio of the training instance.

Table S.15: Area under the ROC curve (AUROC) in the test data for SCRON, RF and XGB across different stages of the conversion funnel

Model	AUROC for the 3 stages of conversion funnel		
	Open	Click	Purchase
SCRON	0.6923	0.6171	0.7248
RF	0.7052	0.6139	0.6916
XGB	0.7084	0.6034	0.6527

The area under the receiver operating characteristic curve (AUROC) for the three different stages of the conversion funnel for RF and XGB models are shown in Table S.15. The RF and XGB models are optimized separately for the three stages. Based on table S.15, we find that the optimally tuned RF and XGB models are performing better than SCRON in the open-stage. However, we find that as one passes through the subsequent stages of the conversion funnel, namely click and purchase, the performance of RF and XGB worsens relative to SCRON; their relative performance decreases as one travels down the funnel. This is not surprising as the RF and XGB being marginal models are not accounting for actions in the previous stage in the funnel. Also, the proportion of cases decreases massively from top to the bottom of the conversion funnel. Our joint modeling approach that borrows strength from user responses across the different stages of the funnel as well as adjusts for severe imbalance by sub-sampling performs considerably better than RF and XGB at the purchase stage. We repeat the profitability analysis conducted in Section 5.5 of the main paper with the RF and XGB models. The results are reported in tables S.16 and S.17 below. We again found that RF and XGB models slightly outperform our proposed joint model in the open stage but are highly sub-optimal for the purchase stage. Relative to SCRON, RF predictive performance was 4% better in the open stage and 37% worse in the purchase stage. Similarly, XGB performed 2% better in the open stage but 27% worse than SCRON in the purchase stage.

Table S.16: Lists of top 5 promotions in high, mid, and low engagement customer segments based on SCRON, Random Forest (RF) and XGBoost (XGB) models

Segment	Promo ID from different models		
	SCRON	RF	XGB
High Engagement	13	17	17
	17	13	13
	22	22	22
	2	5	4
	21	4	5
Mid Engagement	21	13	13
	13	22	22
	22	17	17
	17	25	20
	20	16	21
Low Engagement	21	13	13
	16	22	22
	13	17	21
	20	25	17
	22	21	20

Table S.17: Detailed conversion rates of the different promotions chosen by the SCRON, Random Forest (RF) and XGBoost (XGB) model in the three different consumer segments

Model	Segment	Conversion Rates across the funnel		
		Open	Click	Purchase
SCRON	High	48.25%	44.56%	12.95%
	Mid	45.39%	38.94%	5.32%
	Low	44.55%	41.99%	3.03%
	Overall	47.09%	42.88%	10.17%
RF	High	50.55%	41.41%	8.83%
	Mid	47.42%	42.77%	4.06%
	Low	46.39%	40.14%	3.04%
	Overall	49.16%	41.65%	6.85%
XGB	High	50.55%	41.41%	8.83%
	Mid	45.39%	38.94%	5.32%
	Low	45.62%	39.39%	3.00%
	Overall	48.62%	40.58%	7.40%

S.5 Proof of Proposition 1

The proof follows by checking the conditions of Theorem 2.1 of Dey and Bhattacharya [2016] (henceforth abbreviated as DB16) that ensures geometric convergence of Markov chains with transformation-MCMC modifications (TMCMC). Let $T = (\Theta, B)$ and dimension$(T) = p$. A distribution h is said to be from a super-exponential family if

$$\lim_{||T||\to\infty} ||T||^{-1} \sum_{j=1}^{p} T_j \frac{\partial \log h(T)}{\partial T_j} = -\infty. \tag{2}$$

Theorem 2.1 of DB16 showed that if the target density $h(T)$ is from a super exponential family and does not have contours parallel to $\{T : |T_1| = |T_2| = \ldots = |T_p|\}$ then the corresponding TMCMC chain is geometric ergodic. We next check that our proposal density defined in Section 4 is super-exponential. The prior distribution $g(T)$ used here is continuous and differentiable. Decompose the logarithm of the proposal density as $\log h(T) = l_S(T) + \log g(T)$ where $l_S(T)$ is the log-likelihood based on sample S defined in Equation (8) of the main paper.

As done in Equation (6) of the main paper, we decompose T in Θ, B, and Σ and $\log g(T) = \log g_1(\Theta) + \log g_2(B) + \log g_3(\Sigma)$. Also, note that Θ is actually the conglomeration of $(\boldsymbol{\beta}, \mathbf{m}, \boldsymbol{\nu})$.

Recall, that the parameters are different for the different stages of the conversion funnel. For e.g., $\beta_j^{(l)}$ varied over $l = 1, 2, 3$ stages of the funnel. For convenience, we would use the notation o, c, and p for stages 1, 2 and 3 of the conversion funnel. From its definition in the main paper, it follows that for $a, l \in \{o, c, p\}$:

$$\begin{aligned} \frac{\partial}{\partial \beta_j^{(l)}} \ell_{ik}^{(a)}(T) &= U_{ikj} I\{l = a\}, & \frac{\partial}{\partial \mu_j^{(l)}} \ell_{ik}^{(a)}(T) &= r_{ikj}^{(a)} I\{l = a\}, \text{ and }, \\ \frac{\partial}{\partial \nu_j^{(l)}} \ell_{ik}^{(a)}(T) &= I\{j = i, l = a\}, & \frac{\partial}{\partial b_j^{(l)}} \ell_{ik}^{(a)}(T) &= I\{j = i, l = a\} . \end{aligned} \tag{3}$$

Next, note that by definition Equation (8) it follows that

$$\begin{aligned} -\frac{\partial}{\partial T_j} l_S(T) = & \sum_{(i,k)\in\Omega} \left(1 - o_{ik} - \frac{1}{1 + e^{\ell_{ik}^o}}\right) \frac{\partial \ell_{ik}^o}{\partial T_j} + \sum_{(i,k)\in\Omega_c} \left(1 - c_{ik} - \frac{1}{1 + e^{\ell_{ik}^c}}\right) \frac{\partial \ell_{ik}^c}{\partial T_j} \\ & + \sum_{(i,k)\in\Omega_p} \left(1 - p_{ik} - \frac{1}{1 + e^{\ell_{ik}^p}}\right) \frac{\partial \ell_{ik}^p}{\partial T_j} . \end{aligned} \tag{4}$$

Since we have used independent Gaussian priors with fixed variances for each of the parameters in Θ, we have:

$$-\sum_{T_j \in \Theta} T_j \frac{\partial}{\partial T_j} \log g(T) = \sum_{T_j \in \Theta} w_j T_j^2 \tag{5}$$

where w_j are positive constants. Also, for the consumer specific effects we have:

$$-\frac{\partial}{\partial b_i^{(l)}} \log g(T) = \sum_{j \neq l} \bar{\Sigma}_{lj} b_i^{(j)} + \bar{\Sigma}_{ll} b_i^{(l)} \text{ for } i = 1, \ldots, N \text{ where } \bar{\Sigma} = \Sigma^{-1}. \tag{6}$$

Finally, note that for the $\bar{\Sigma}$ parameter, the contributions are from g_2 and g_3:

$$\frac{\partial}{\partial \bar{\Sigma}} \log g(T) = \frac{\partial}{\partial \bar{\Sigma}} \log g_2(T) + \frac{\partial}{\partial \bar{\Sigma}} \log g_3(T).$$

The first term is the derivative with respect to $\bar{\Sigma}$ of multivariate normal density of B. The second term is from the Wishart prior $\log g_3(T) = (d+4)/2 \log |\bar{\Sigma}| - \text{tr}(\bar{\Sigma})/2 + \text{constant}$ where d is the degrees of freedom of the prior. Here, the covariance of the prior is set to I. Thus,

$$-\frac{\partial}{\partial \bar{\Sigma}} \log g(T) = -(N + d/2 + 2)\bar{\Sigma} + 2^{-1} \sum_{i=1}^{N} \boldsymbol{b}_i \boldsymbol{b}_i' \ . \tag{7}$$

Now, from Equation (4) it follows that $\sum_{j=1}^{p} T_j \frac{\partial}{\partial T_j} l_S(T)$ is bounded above by

$$\sum_{(i,k) \in \Omega} o_{ik} T_j \frac{\partial \ell_{ik}^o}{\partial T_j} + \sum_{(i,k) \in \Omega_c} c_{ik} T_j \frac{\partial \ell_{ik}^c}{\partial T_j} + \sum_{(i,k) \in \Omega_p} p_{ik} T_j \frac{\partial \ell_{ik}^p}{\partial T_j}.$$

From Equation (3) it follows that the above does not involve any quadratic terms of the parameters. Equations (5)–(7) show that all the parameters that appear in the above sum are also involved quadratically in $\sum_{j=1}^{p} T_j \frac{\partial}{\partial T_j} \log g(T)$. Thus, for any fixed N, $\sum_{j=1}^{p} T_j \frac{\partial}{\partial T_j} l_S(T)$ is dominated by $\sum_{j=1}^{p} T_j \frac{\partial}{\partial T_j} \log g(T)$. Note, both assumptions A1 and A2 are essential to ensure this domination. Moreover, the quadratic terms have negative coefficients in $\sum_{j=1}^{p} T_j \frac{\partial}{\partial T_j} \log g(T)$ and therefore, we arrive at Equation (2) which completes the proof.

S.6 Discussion on Predictive Optimality of BMHT

The adjustments, prescribed in Section 4.2.3 of the main paper, results in consistent estimates of the model parameters. Based on this consistent-estimation scheme, our proposed

method can be used for predicting the response of future customers. In Section 5 we have provided the ROC curves for test data sets based on models estimated by our proposed BMHT procedure. Following the analysis framework developed in Fithian and Hastie [2014], asymptotic optimality of the BMHT methodology can be established. For any fixed covariate matrix X_{te}, let $Y_{\mathsf{te}} = \{(O_{ik}, C_{ik}, P_{ik}) : i = 1, \ldots, M; k = 1, \ldots, K\}$ responses from m new customers generated from Equations (1)-(4) with true parameters being Θ_0 and $B_{\mathsf{te}} = \{b_i^{\mathsf{te}} : i = 1, \ldots, M\}$ being i.i.d. from $N_3(0, \Sigma_0)$, for any positive definite Σ_0. Denote the true distribution of Y_{te} by $q(Y_{\mathsf{te}}|X_{\mathsf{te}}, \Theta_0, B_{\mathsf{te}})$.

Let X_{tr} be the fixed training matrix and Y_{tr} is generated from (1)-(4) using X_{tr}, Θ_0 and $B_{\mathsf{tr}} = \{b_i^{\mathsf{tr}} : i = 1, \ldots, N\}$ being i.i.d from $N_3(0, \Sigma_0)$. Consider the plugin estimate $q(Y_{\mathsf{te}}|X_{\mathsf{te}}, \hat{\Theta}^S, \hat{B}^S)$ for the predictive distribution Y_{te} where $\hat{B}^S$ contains i.i.d. vectors from $N(0, \hat{\Sigma}^S)$. Note, that $\hat{\Theta}^S, \hat{B}^S$ are functions of $(Y_{\mathsf{tr}}, X_{\mathsf{tr}})$. The analysis in Fithian and Hastie [2014] can be extented to provide weak consistency of the estimated paramters $\hat{\Theta}^S \xrightarrow{P} \Theta_0$ and $\hat{\Sigma}^S \xrightarrow{P} \Sigma_0$. These when substituted in (5) yield the following result on good asymptotic performance by the BMHT based plugin predictive distribution.

Proposition 1. *For any fixed K, as $M, N \to \infty$ the plugin predictive distribution using BMHT estimates from a subsample S that satisfies assumptions A1 and A2 converges to the true distribution in Kullback-Leibler loss:*

$$\frac{1}{M} \int \int q(Y_{te}|X_{te}, \Theta_0, B_{te}) \left\{ \log \frac{q(Y_{te}|X_{te}, \hat{\Theta}^S, \hat{\Sigma}^S)}{q(Y_{te}|X_{te}, \Theta_0, B_{te})} \right\} dY_{te}\, dY_{tr} \xrightarrow{P} 0.$$

References

J. Cohen. *Statistical power analysis for the behavioral sciences.* Academic press, 2013.

K. K. Dey and S. Bhattacharya. On geometric ergodicity of additive and multiplicative transformation-based markov chain monte carlo in high dimensions. *Brazilian Journal of Probability and Statistics*, 30(4):570–613, 2016.

W. Fithian and T. Hastie. Local case-control sampling: Efficient subsampling in imbalanced data sets. *Annals of Statistics*, 42(5):1693, 2014.

G. Malsiner-Walli and H. Wagner. Comparing spike and slab priors for bayesian variable selection. *Austrian Journal of Statistics*, 2011.

D. F. McCaffrey, B. A. Griffin, D. Almirall, M. E. Slaughter, R. Ramchand, and L. F. Burgette. A tutorial on propensity score estimation for multiple treatments using generalized boosted models. *Statistics in medicine*, 32(19):3388–3414, 2013.